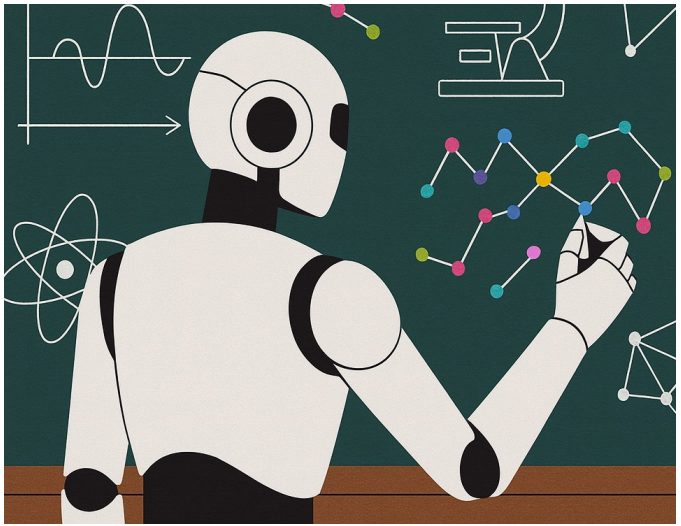
Winter semester 2025/26

AI for Quantum Physics

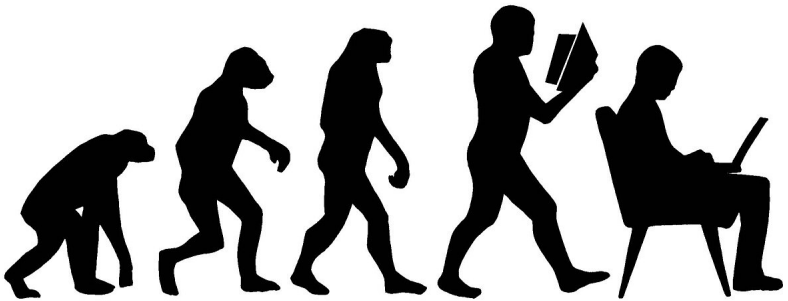


Xuemei Gu
xuemei.gu@uni-jena.de

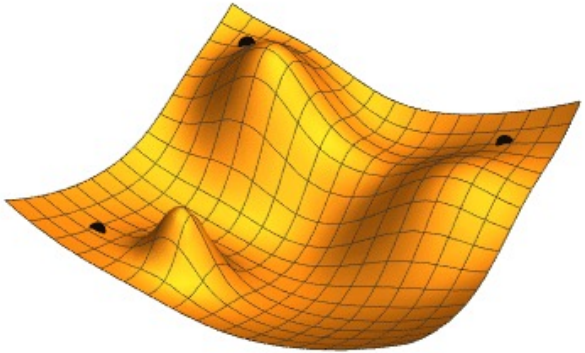
If you have any questions, feel free to reach out!

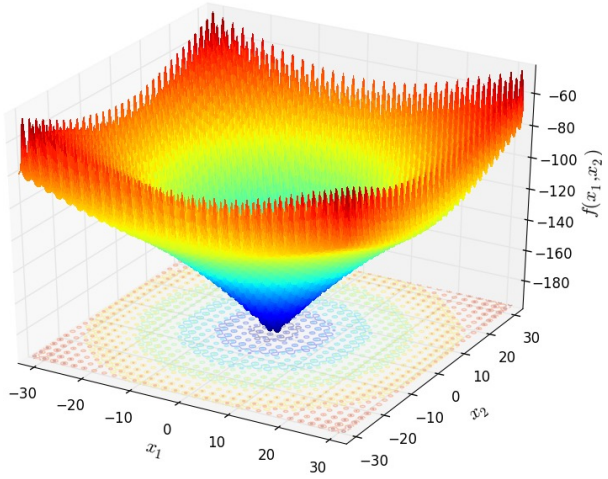
Today's topic:

- What are evolutionary algorithms?
- How are they implemented?
- Applications in (Quantum) Physics

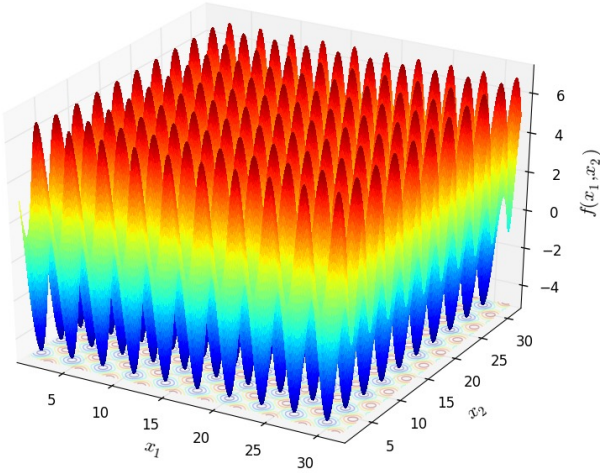


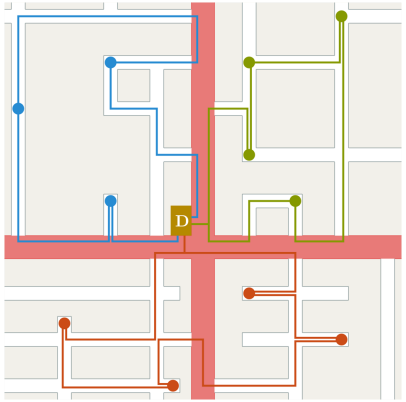
Gradient descent?





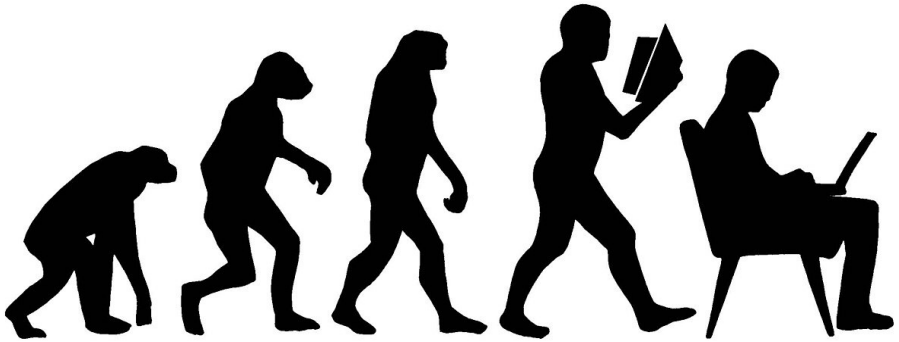






Evolutionary algorithms

a nature inspired method to optimization



Adaption is intelligence

Survival of the “fittest”

The algorithms mimic biological evolution, using **variation, selection, and reproduction**, to explore complex search spaces without gradients or explicit formulas

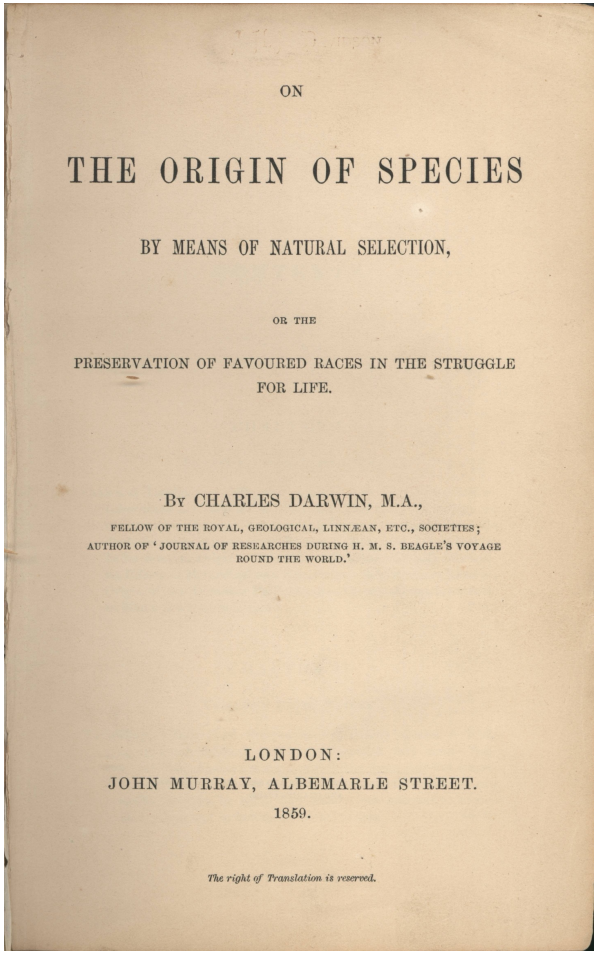
Evolution and Nature Selection

Darwin's Principles of Evolution

Variation: Individuals in a population differ in traits such as behavior or appearance.

Inheritance: Traits are passed from parents to offspring, making them resemble their parents more than unrelated individuals.

Selection: Individuals better adapted to their environment survive and reproduce more successfully.



ON

THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE

PRESERVATION OF FAVOURED RACES IN THE STRUGGLE FOR LIFE.

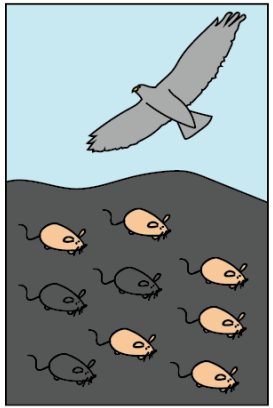
By CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNÆAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE ROUND THE WORLD.'

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1859.

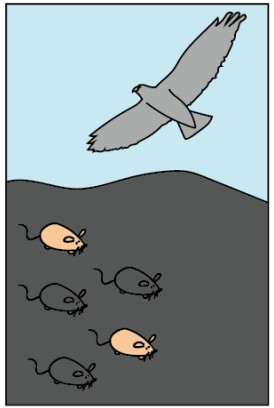
The right of Translation is reserved.

Evolution and Nature Selection



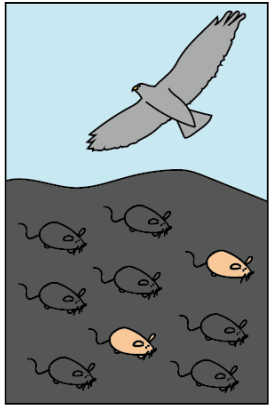
A population of mice has moved into a new area where the rocks are very dark. Due to natural genetic variation, some mice are black, while others are tan.

Some mice are eaten by birds

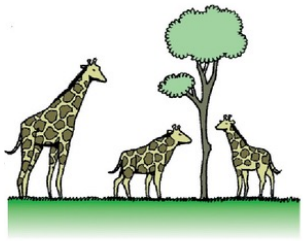


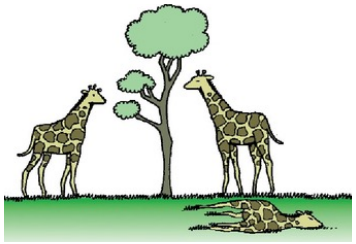
Tan mice are more visible to predatory birds than black mice. Thus, tan mice are eaten at higher frequency than black mice. Only the surviving mice reach reproductive age and leave offspring.

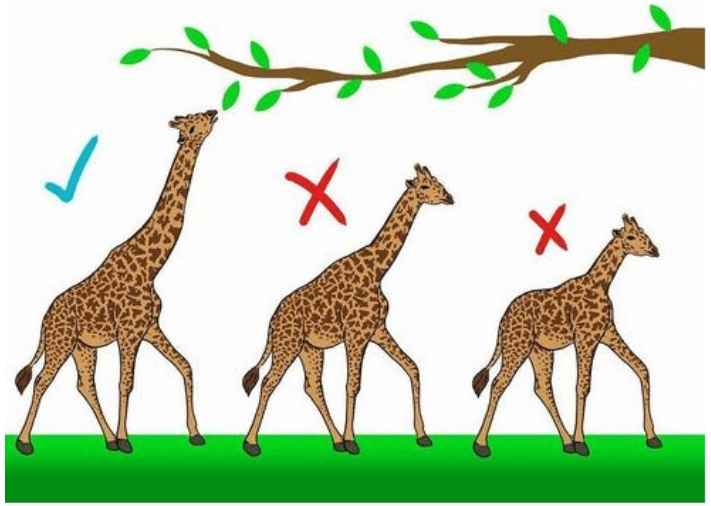
Mice reproduce, giving next generation



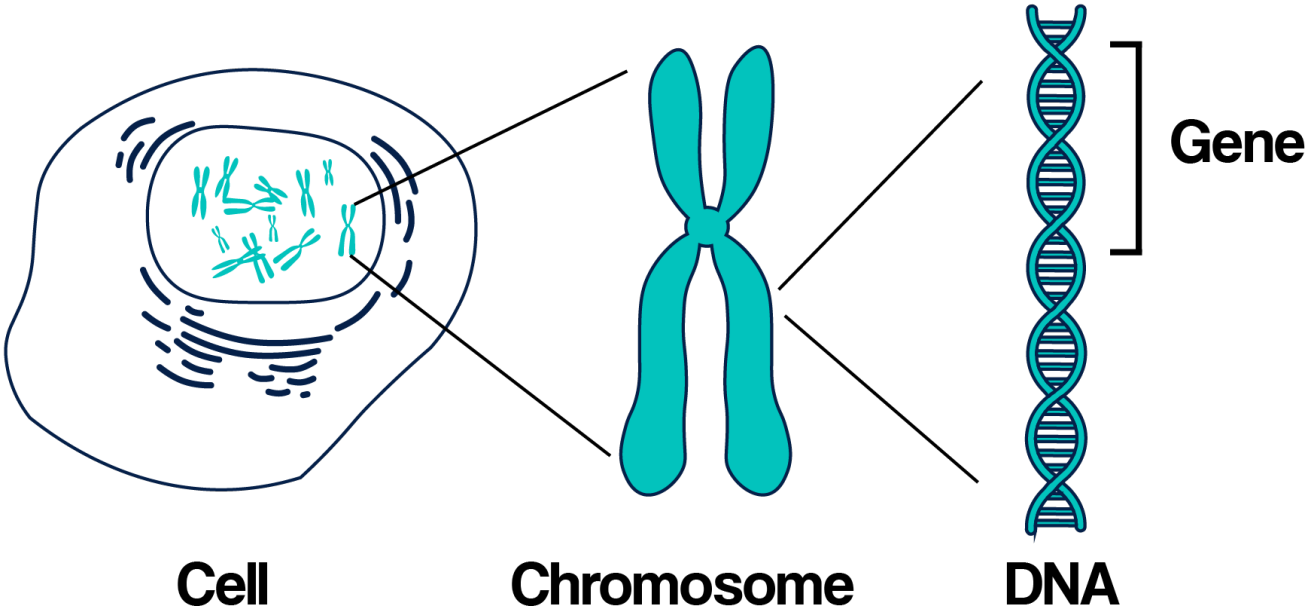
Because black mice had a higher chance of leaving offspring than tan mice, the next generation contains a higher fraction of black mice than the previous generation.



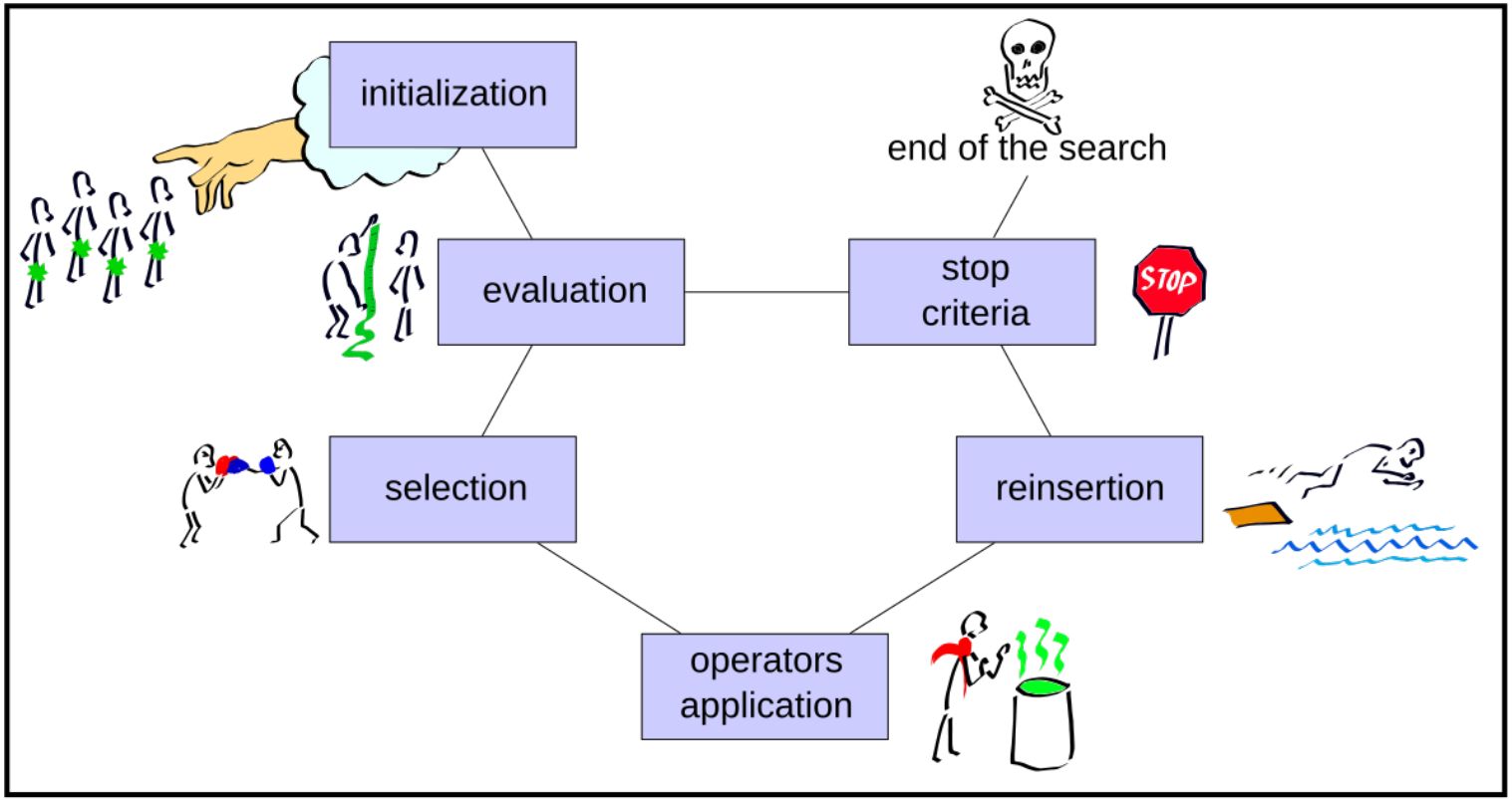




How does evolutionary algorithms work?



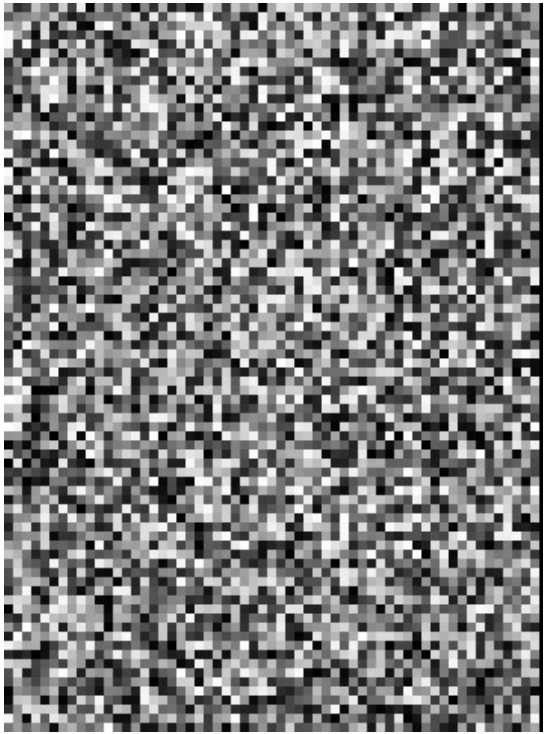
How does evolutionary algorithms work?



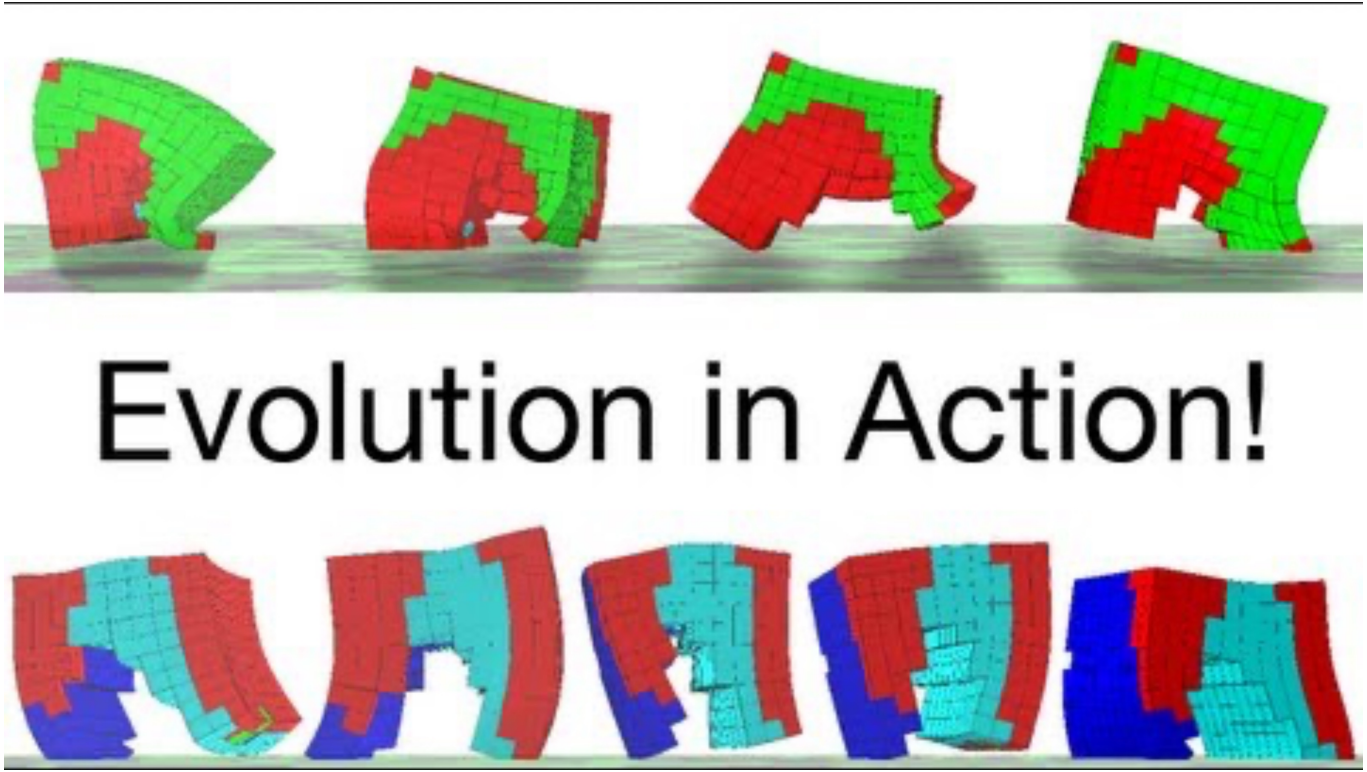
General schema of an Evolutionary Algorithm (EA)

Evolutionary algorithms

- Genetic Algorithm (GA)
- Genetic Programming (GP)
- Differential Evolution (DE)
- Evolution Strategy (ES)
- Evolutionary Programming (EP)
- …

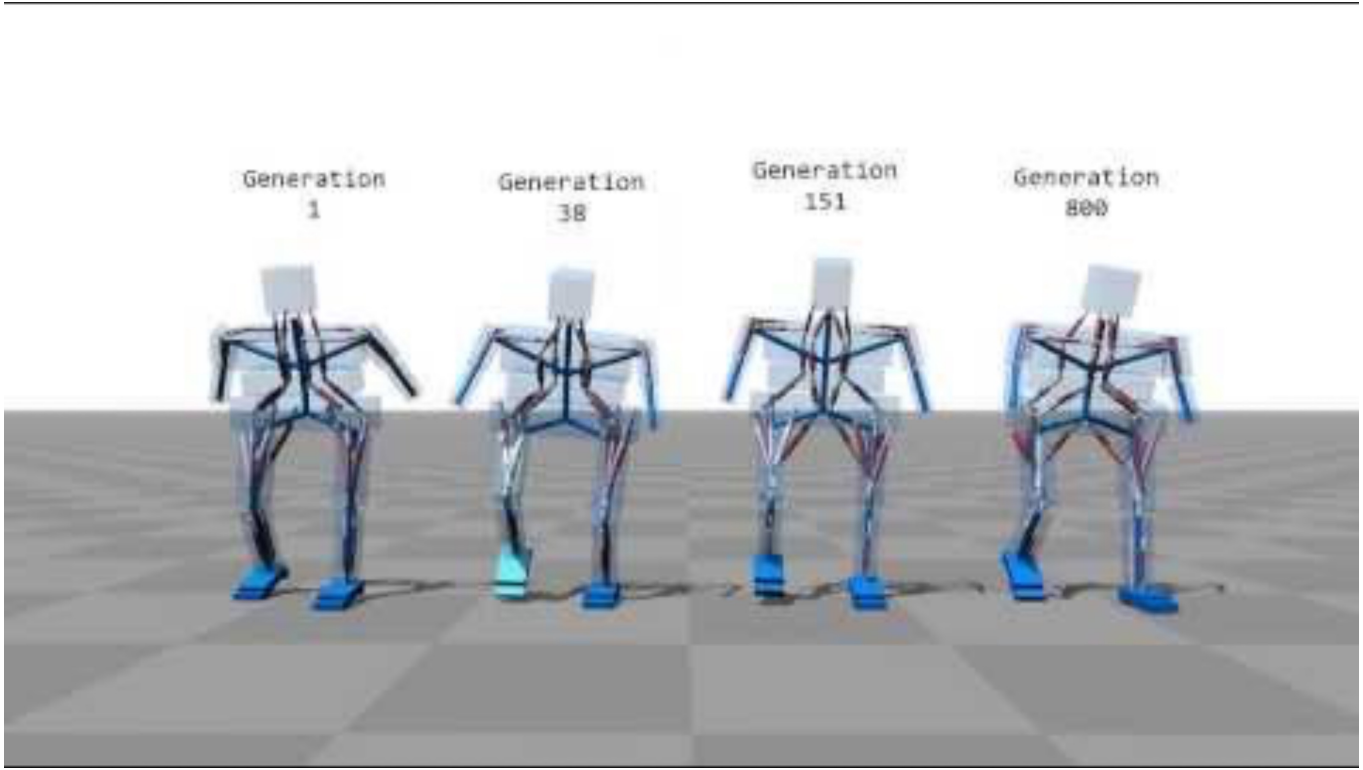


Evolving Soft Robots



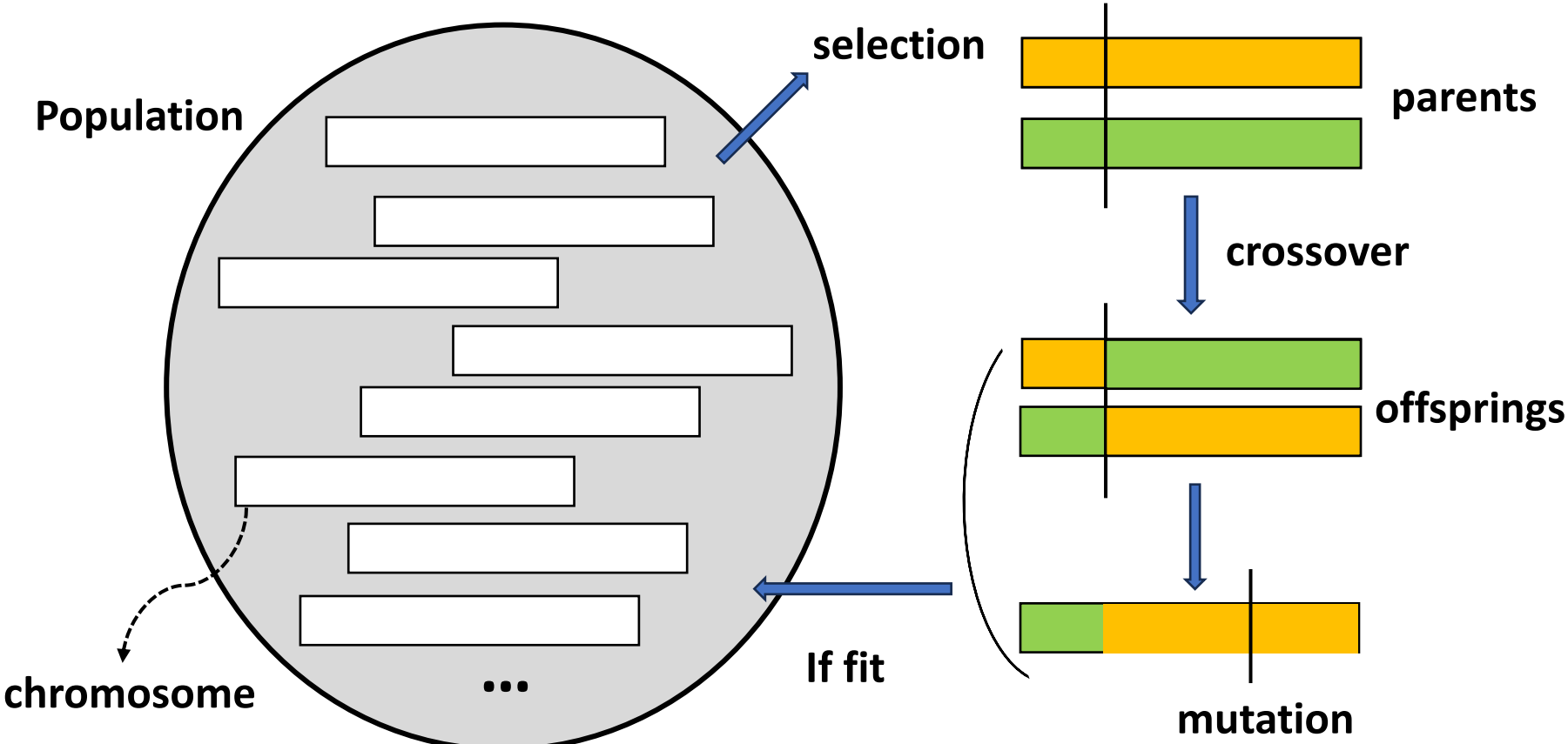
https://www.youtube.com/watch?v=z9ptOeByLA4

Learning to Walk



https://www.youtube.com/watch?v=GqVuV7Khiag

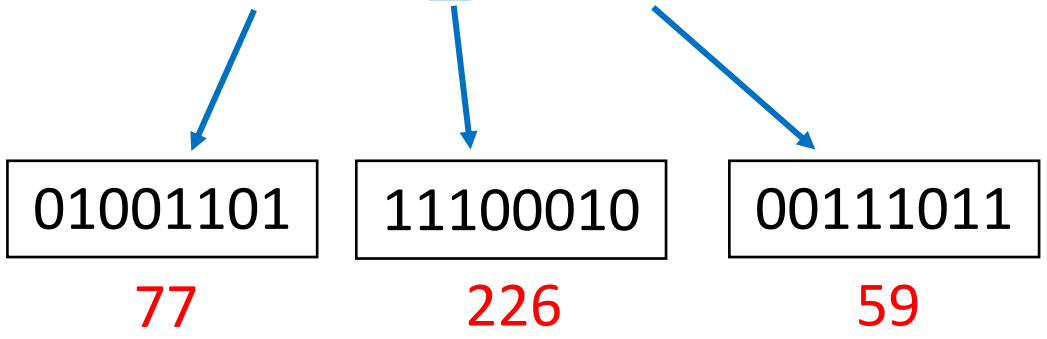
Genetic algorithm (GA)


Population
selection
parents
crossover
offsprings
mutation
If fit
chromosome
...

How to apply this idea to real-valued optimization problems?

$$f(x,y,z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \qquad x,y,z \in [-3,3]$$

As an example, we assume each variable stored as 8 bit (binary)



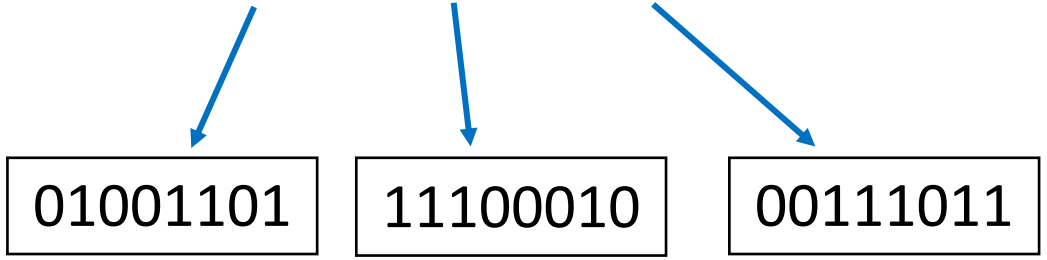
$$x \text{ (or } y \text{ or } z) = -3 + \frac{\text{integer}}{255} \times 6$$

(x,y,z) ≈ (−1.19, 2.32, −1.61)

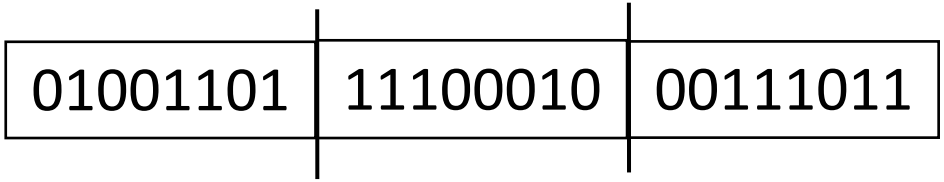
How to apply this idea to real-valued optimization problems?

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \qquad x, y, z \in [-3, 3]$$

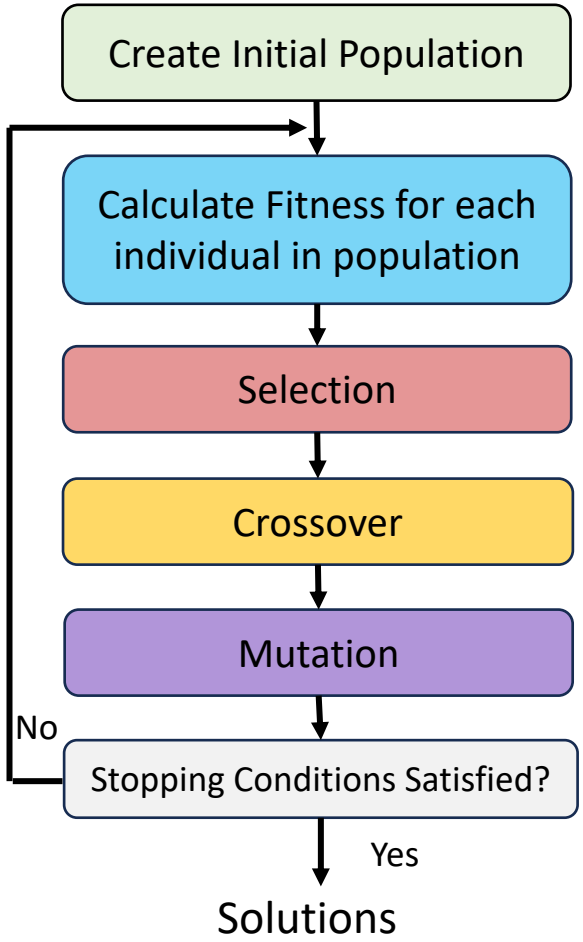
As an example, we assume each variable stored as 8 bit (binary)



chromosome

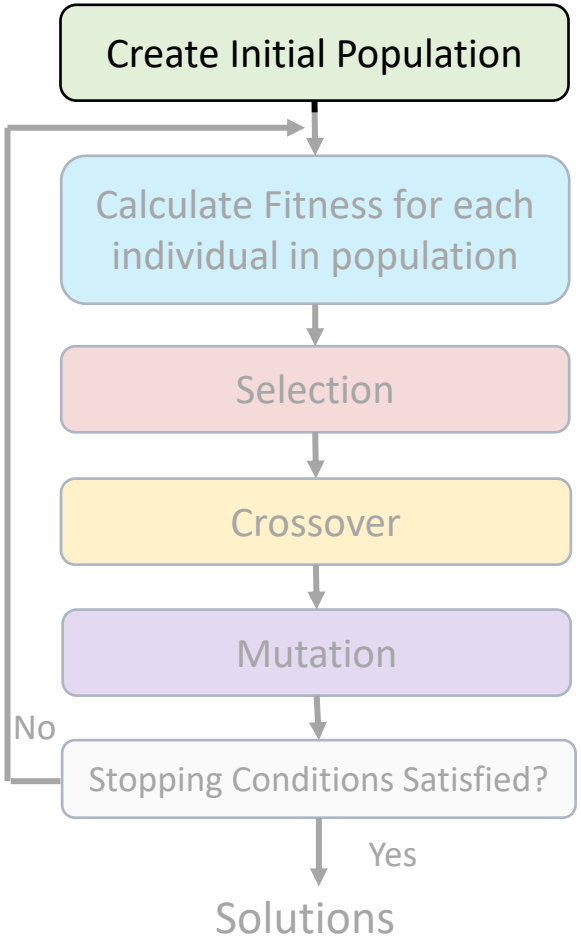


How to apply?

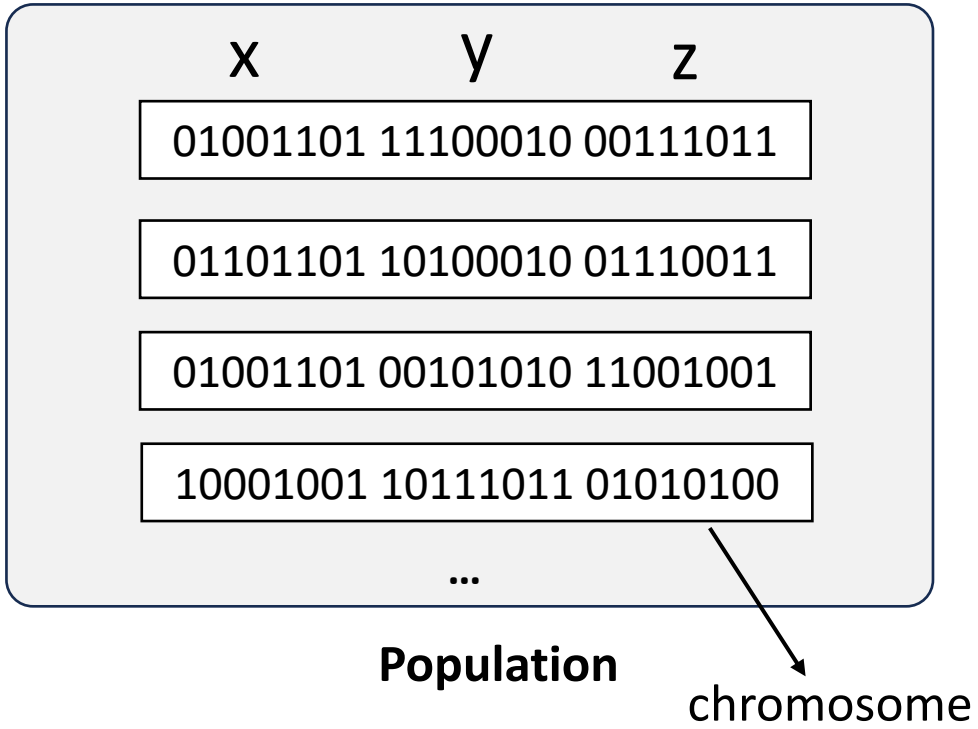



```
initialize population;
evaluate population;
while Termination Criteria Not Satisfied
{
    select parents for reproduction;
    perform crossover and mutation;
    evaluate population;
}
```

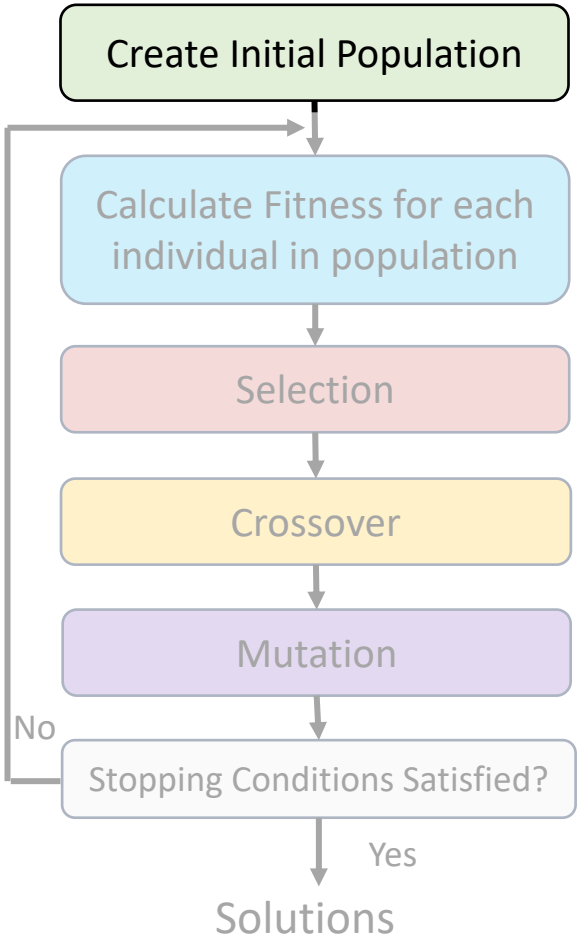
How to apply?



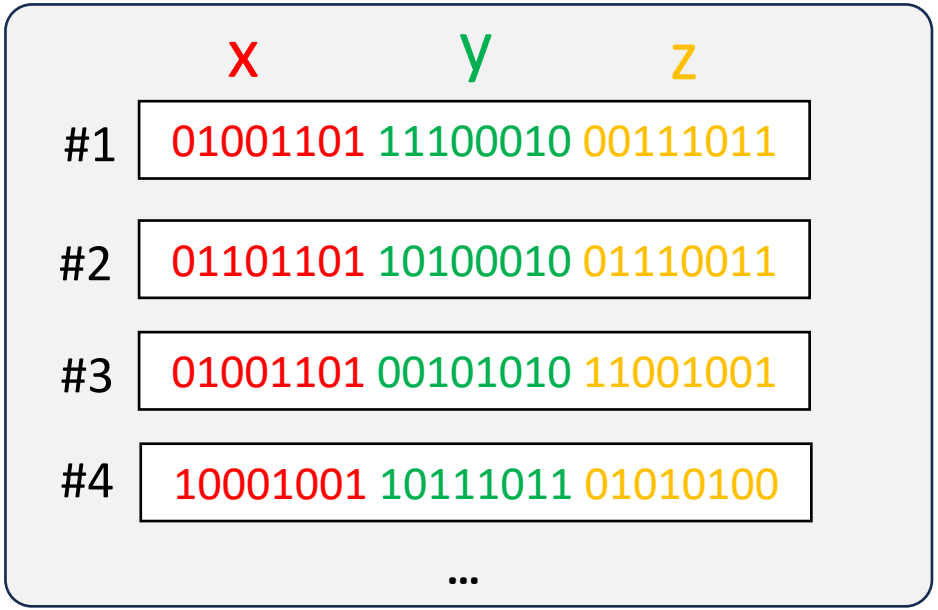
$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$



How to apply?



$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$



Population

How to apply?

Create Initial Population

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

Population Size

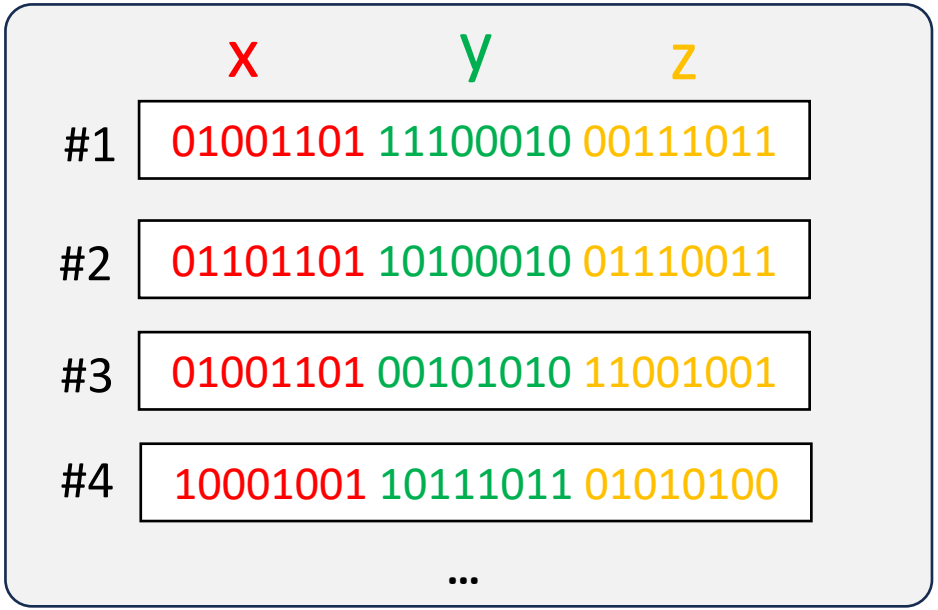
How many chromosomes in one generation

Too many:

extremely slow

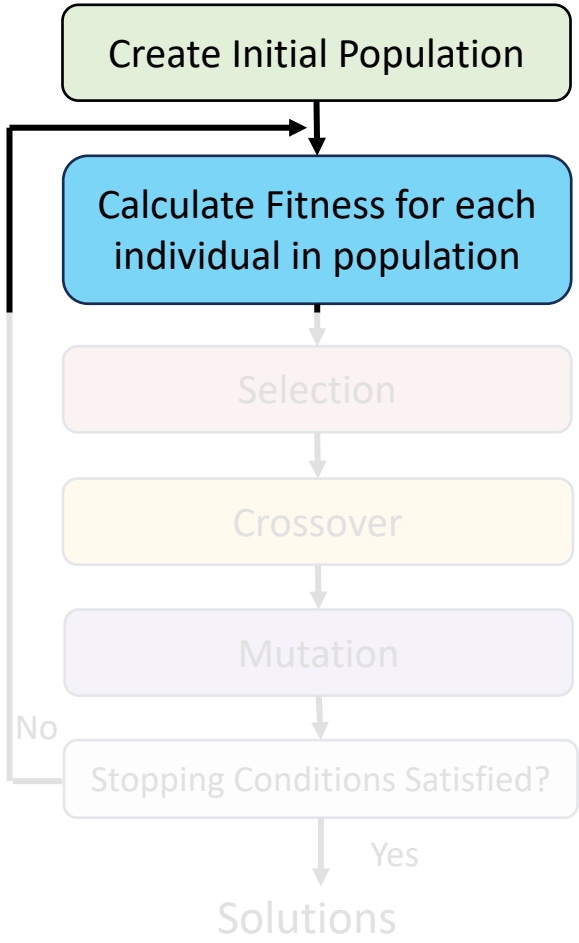
Too few:

not many possibilities for mating (crossover), only a part of the search space will be sampled



Population

How to apply?



$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

	x	y	z
#1	01001101	11100010	00111011
#2	01101101	10100010	01110011
#3	01001101	00101010	11001001
#4	10001001	10111011	01010100
	...		

Population

Calculate Fitness for each individual in population

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \qquad x, y, z \in [-3, 3]$$

	x	y	z
#1	01001101	11100010	00111011
#2	01101101	10100010	01110011
#3	01001101	00101010	11001001
#4	10001001	10111011	01010100
	...		

(77, 226, 59)

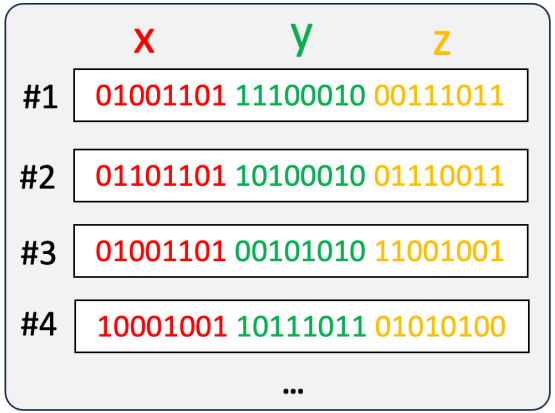
Decode: $x \text{ (or } y \text{ or } z) = -3 + \dfrac{\text{integer}}{255} \times 6$

$$x = -3 + \tfrac{6}{255} \cdot 77 = -1.188235$$
$$y = -3 + \tfrac{6}{255} \cdot 226 = 2.317647$$
$$z = -3 + \tfrac{6}{255} \cdot 59 = -1.611765$$

(-1.188235, 2.317647, -1.611765)

Calculate Fitness for each individual in population

$$f(x,y,z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \qquad x,y,z \in [-3,3]$$



Fitness when we minimize f(x,y,z)

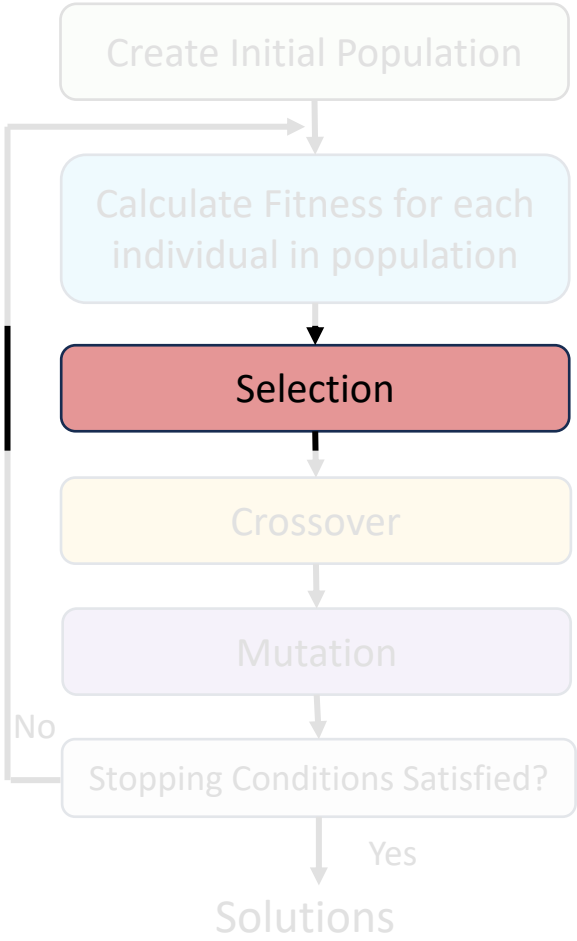
Sign flip: $\text{fitness} = -f$

Shifted for roulette: $\text{fitness}_i = f_{max} - f_i + \varepsilon$

Fitness when we maximize f(x,y,z): $\text{fitness} = f$

#1	**(77, 226, 59)**	(-1.188235, 2.317647, -1.611765)	f(x,y,z)=9.445288093
#2	**(109,162,115)**	(-0.435294, 0.811765, -0.294118)	f(x,y,z)= 0.518914388
#3	**(77, 42, 201)**	(−1.188235, −2.011765, 1.729412)	f(x,y,z)= 8.147812489
#4	**(137,187,84)**	(0.223529, 1.400000, −1.023529)	f(x,y,z)= 3.397275873

How to apply?



$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

#1	01001101 11100010 00111011	f(x,y,z)=9.445288093
#2	01101101 10100010 01110011	f(x,y,z)= 0.518914388
#3	01001101 00101010 11001001	f(x,y,z)= 8.147812489
#4	10001001 10111011 01010100	f(x,y,z)= 3.397275873
	...	

Let's say we maximize f(x,y,z): $\text{fitness} = f$

Straightforward selection (high fitness)

#1	01001101 11100010 00111011	**parents**
#3	01001101 00101010 11001001	

Selection

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

Let's say we maximize f(x,y,z): $\text{fitness} = f$

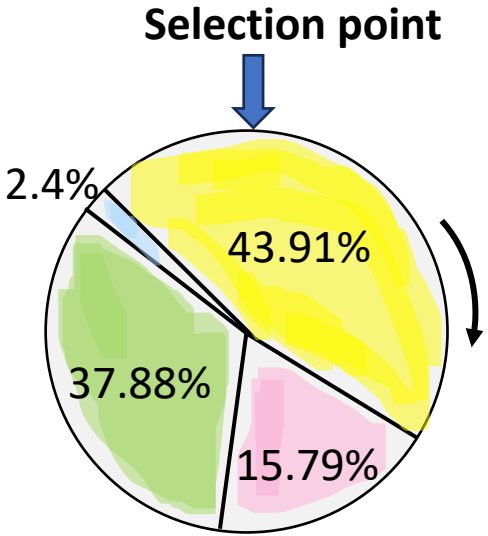
#1	01001101 11100010 00111011
#2	01101101 10100010 01110011
#3	01001101 00101010 11001001
#4	10001001 10111011 01010100
	...

f1: f(x,y,z)=9.445288093

f2: f(x,y,z)= 0.518914388

f3: f(x,y,z)= 8.147812489

f4: f(x,y,z)= 3.397275873



spinning wheel selection

f1/(f1+f2+f3+f4)≈0.439126

f2/(f1+f2+f3+f4)≈0.0241251

f3/(f1+f2+f3+f4)≈0.378804

f4/(f1+f2+f3+f4)≈0.157945

Selection

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

Let's say we maximize f(x,y,z): $\text{fitness} = f$

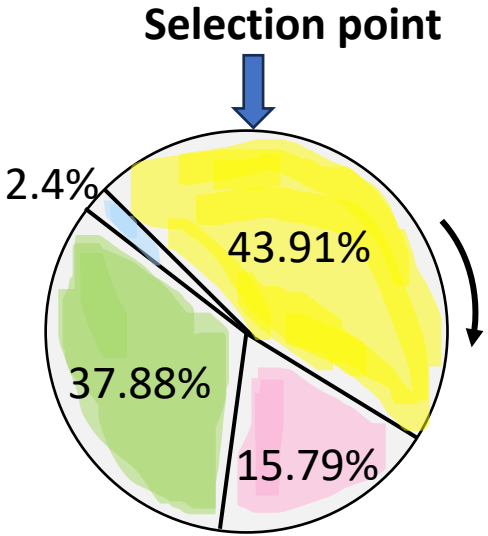
#1	01001101 11100010 00111011
#2	01101101 10100010 01110011
#3	01001101 00101010 11001001
#4	10001001 10111011 01010100
	...

f1: f(x,y,z)=9.445288093

f2: f(x,y,z)= 0.518914388

f3: f(x,y,z)= 8.147812489

f4: f(x,y,z)= 3.397275873



spinning wheel selection

#1	01001101 11100010 00111011
#3	01001101 00101010 11001001

parents

Selection

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

Let's say we maximize f(x,y,z): $\text{fitness} = f$

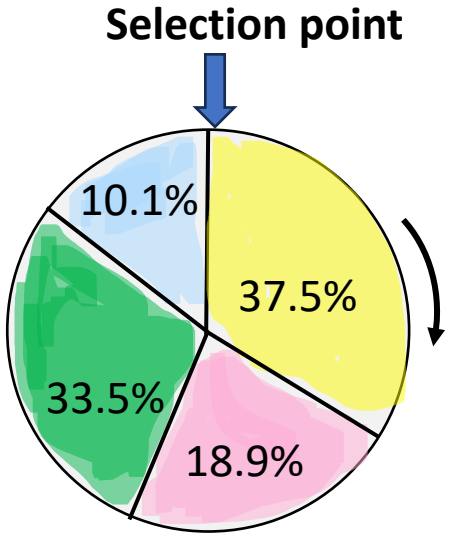
#1	01001101 11100010 00111011
#2	01101101 10100010 01110011
#3	01001101 00101010 11001001
#4	10001001 10111011 01010100
	...

f1: f(x,y,z)=9.445288093

f2: f(x,y,z)= 0.518914388

f3: f(x,y,z)= 8.147812489

f4: f(x,y,z)= 3.397275873



fitness scaling Linear scale: Scaled fitness=a x (raw fitness) +b

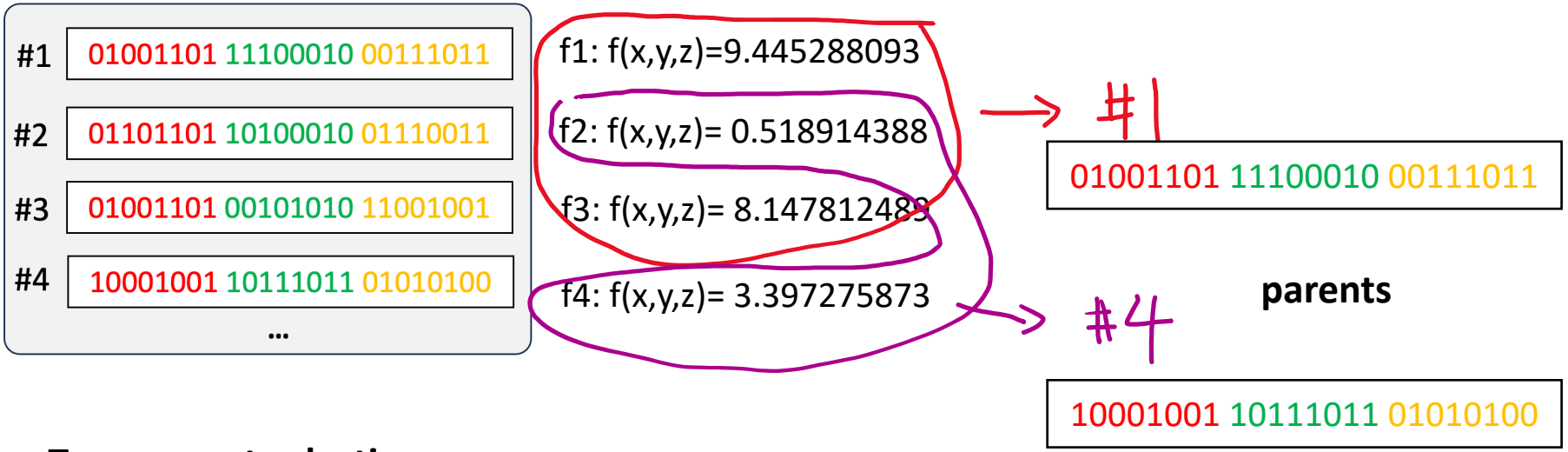
$$f_i' = 0.661\, f_i + 1.827$$

Raw (fi)	Scaled (fi)	Probability
9.4453	8.066	0.375
0.5189	2.169	0.101
8.1478	7.217	0.335
3.3973	4.072	0.189

Selection

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z)$$

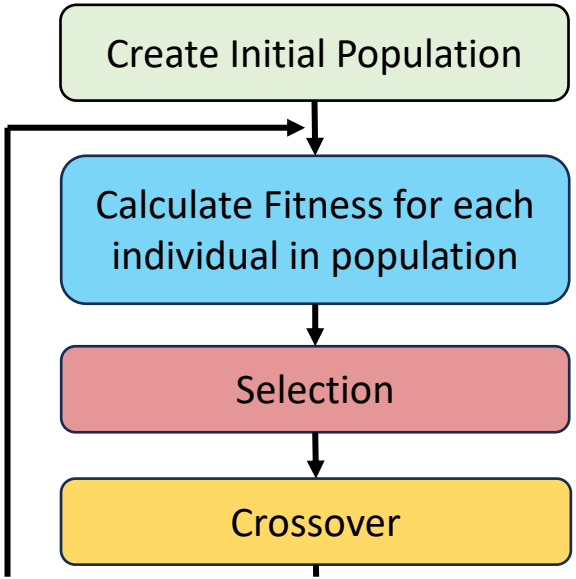
Let's say we maximize f(x,y,z): $\text{fitness} = f$



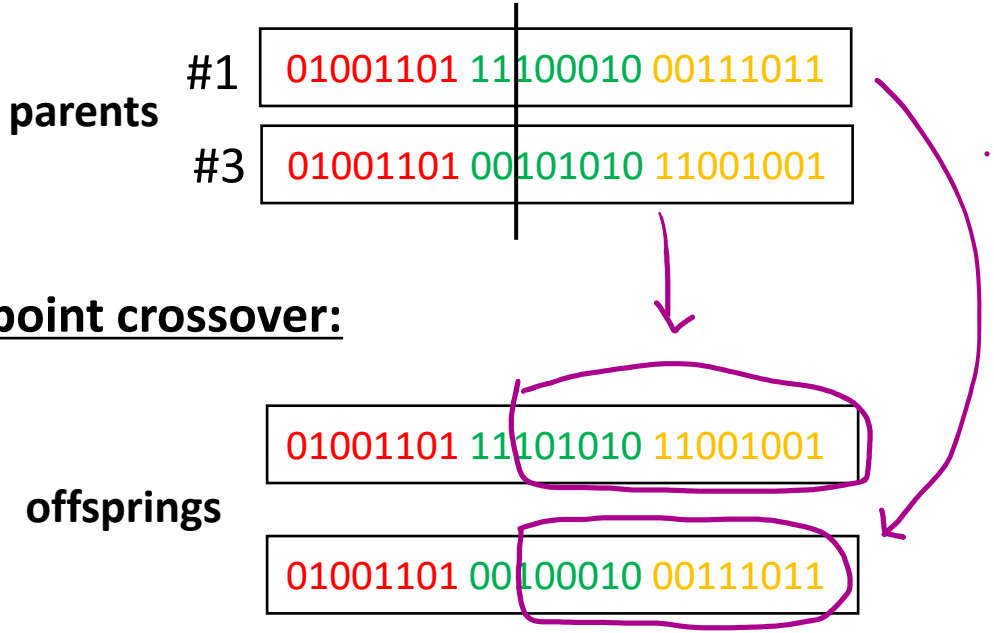
Tournament selection

In each round of the tournament selection, two or more individuals are randomly picked from the population, and the one with the highest fitness score wins and get selected.

How to apply?

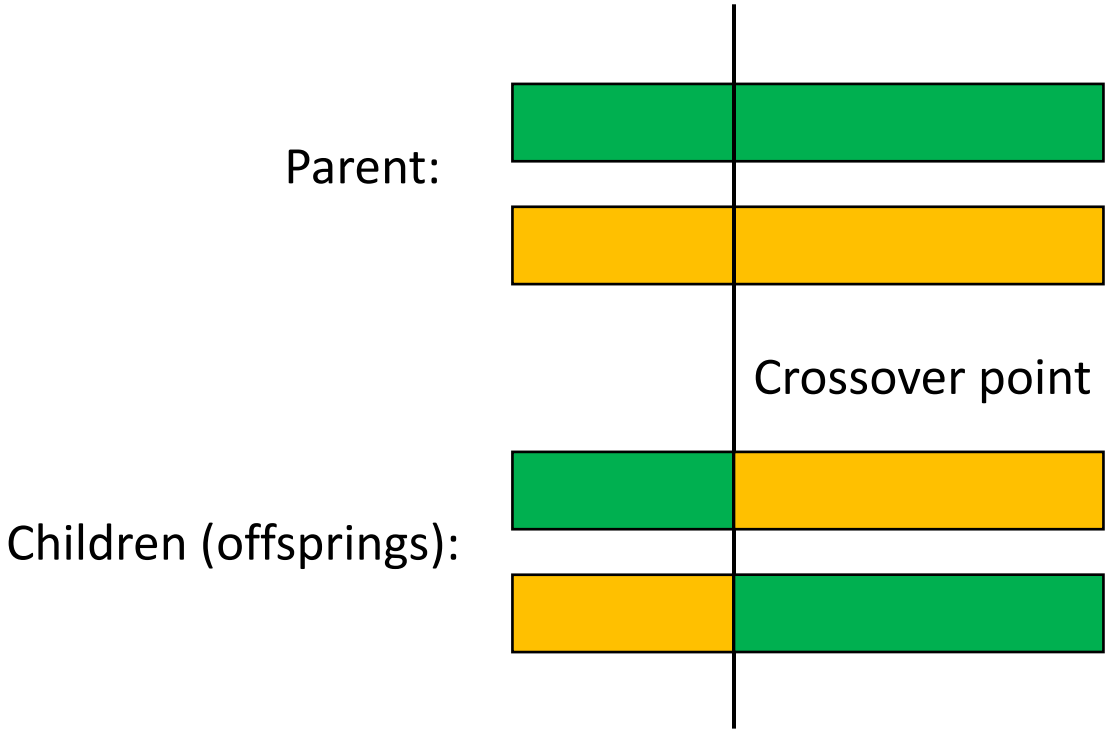


Selection e.g.,

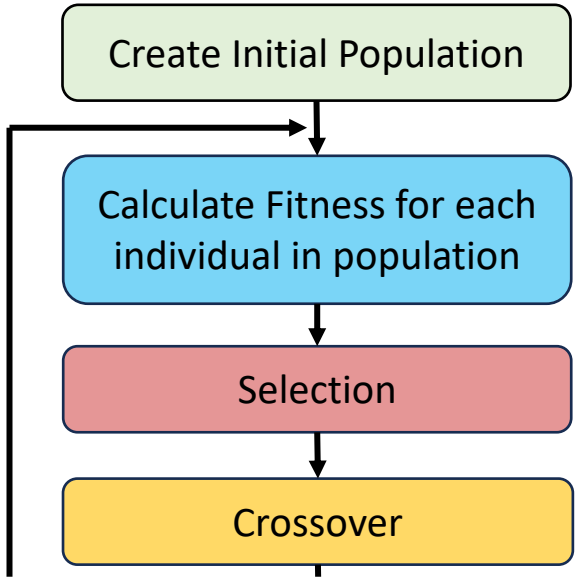


A location on the chromosomes of both parents is selected randomly. The location is referred to as the crossover point or cut point. As a result, we get two offspring, where each of them carries some genetic information from both parents.

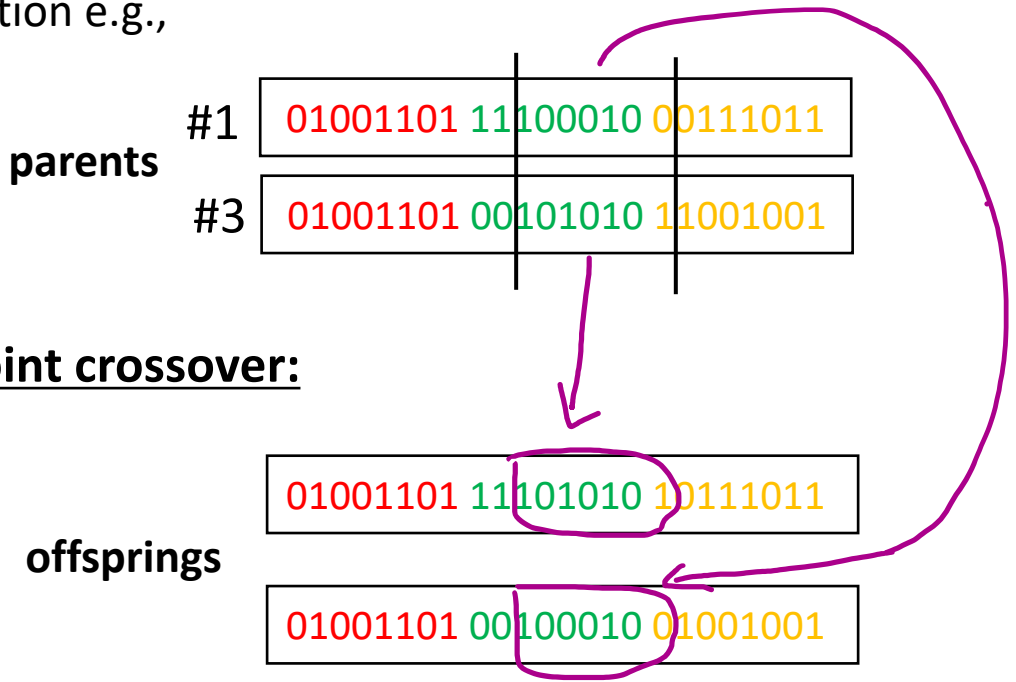
single point crossover:



How to apply?

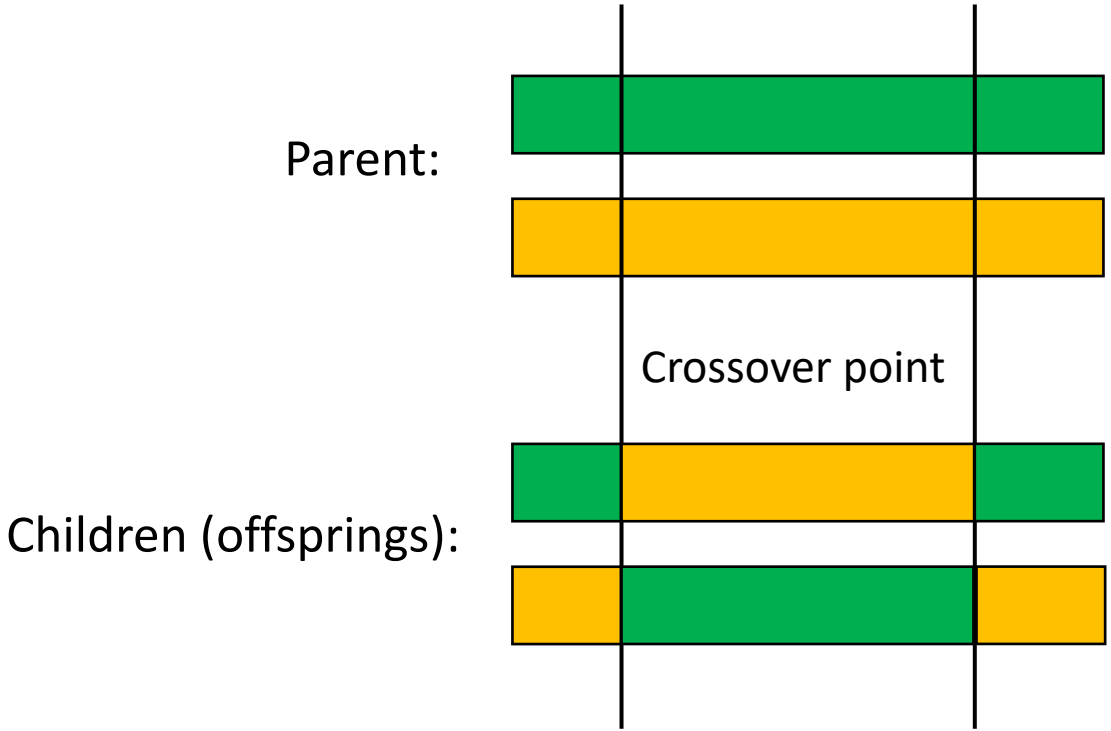


Selection e.g.,

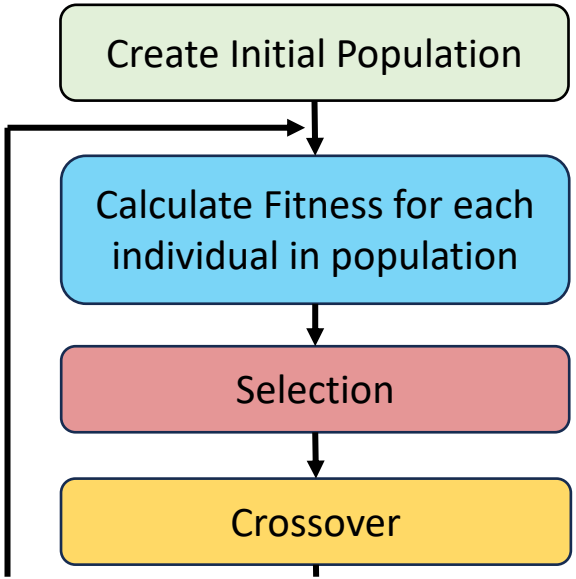


Two locations on the chromosomes of both parents are selected randomly. The genes residing between these points are swapped between the two parent chromosomes. The two point crossover method can be implemented by carrying out two single-point crossovers, each with a different crossover point.

two point crossover:



How to apply?



Selection e.g.,

parents

#1 01001101 11100010 00111011

#3 01001101 00101010 11001001

uniform crossover:

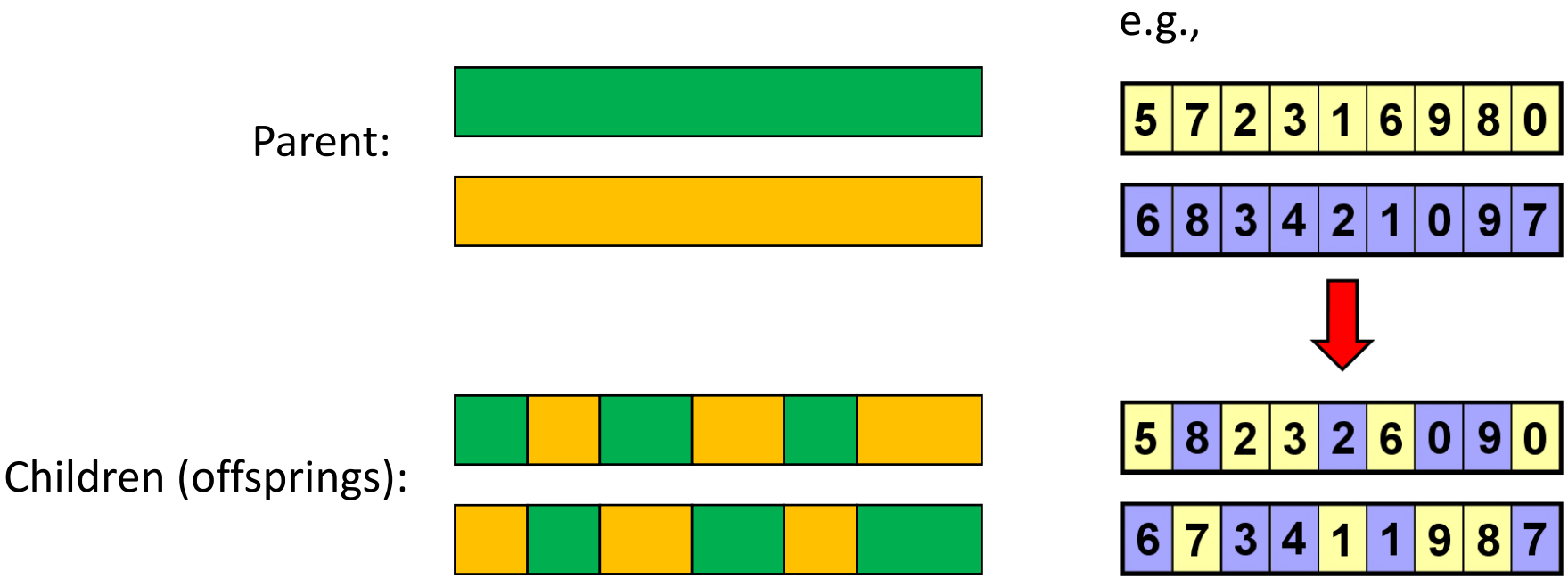
offsprings

01001101 10100010 01011001

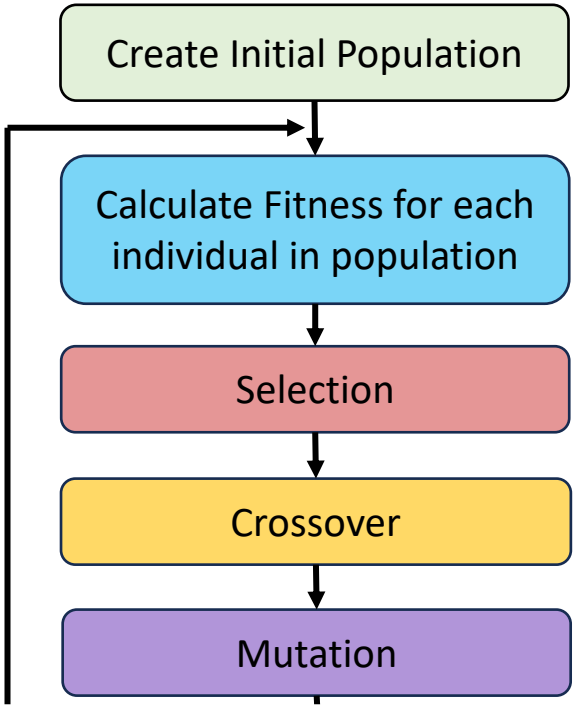
01001101 01101010 10101011

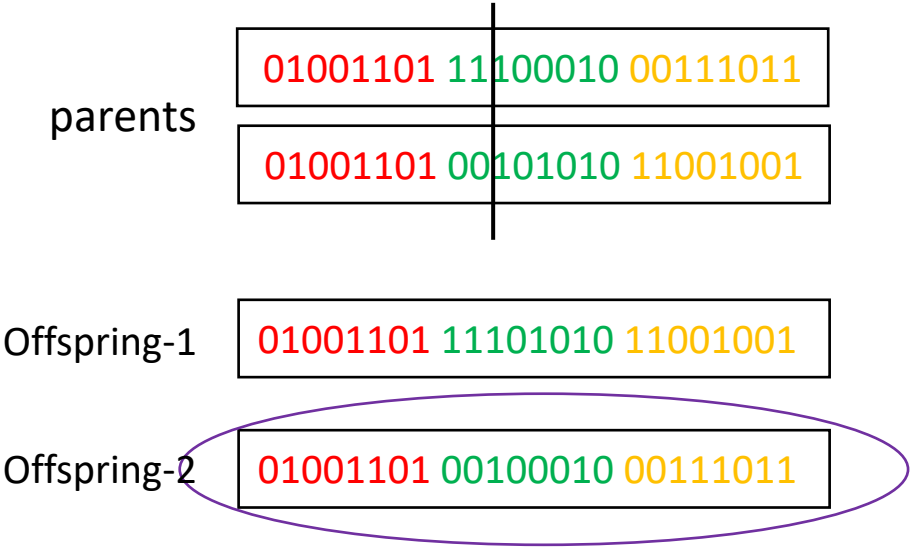
Each gene is independently determined by randomly choosing one of the parents. When the random distribution is 50%, each parent has the same chance of influencing the offsprings

uniform crossover:

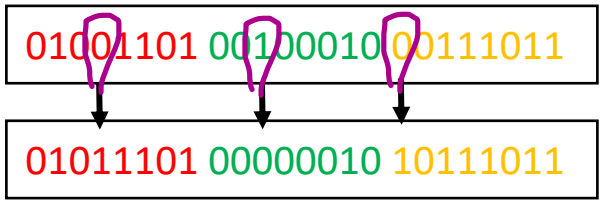


How to apply?

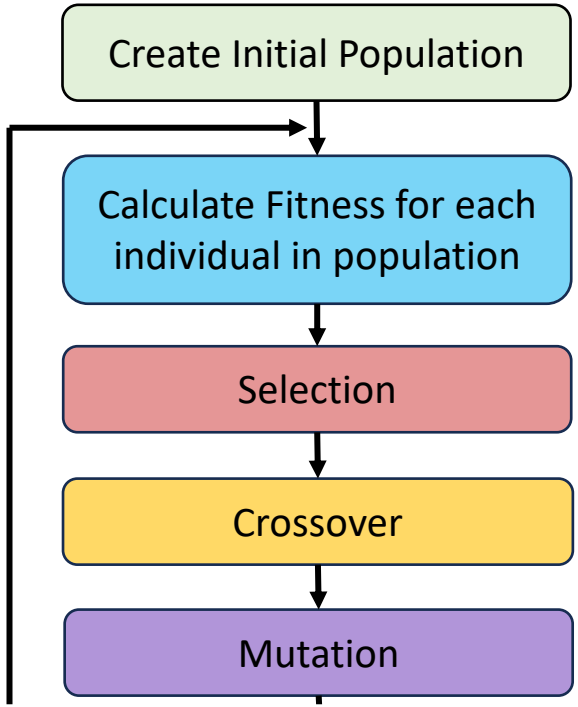


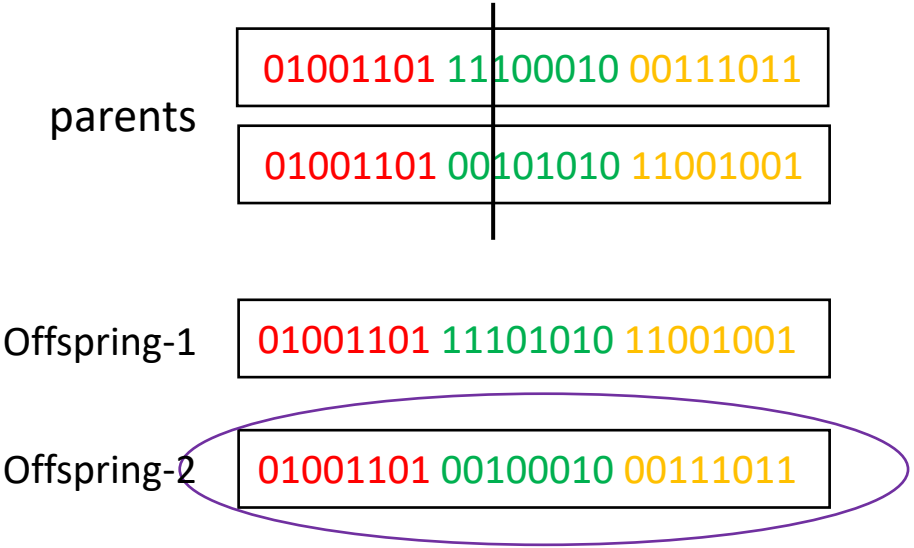


Flip bit mutation

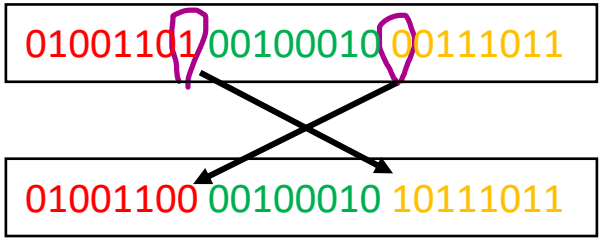


How to apply?

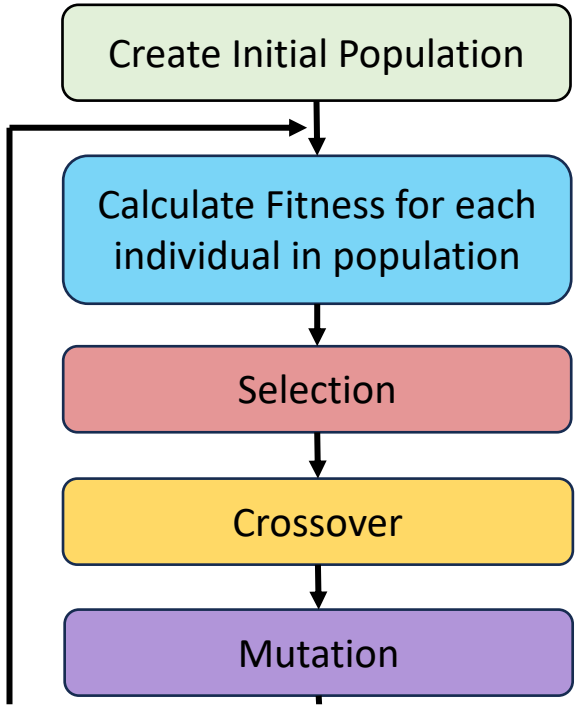


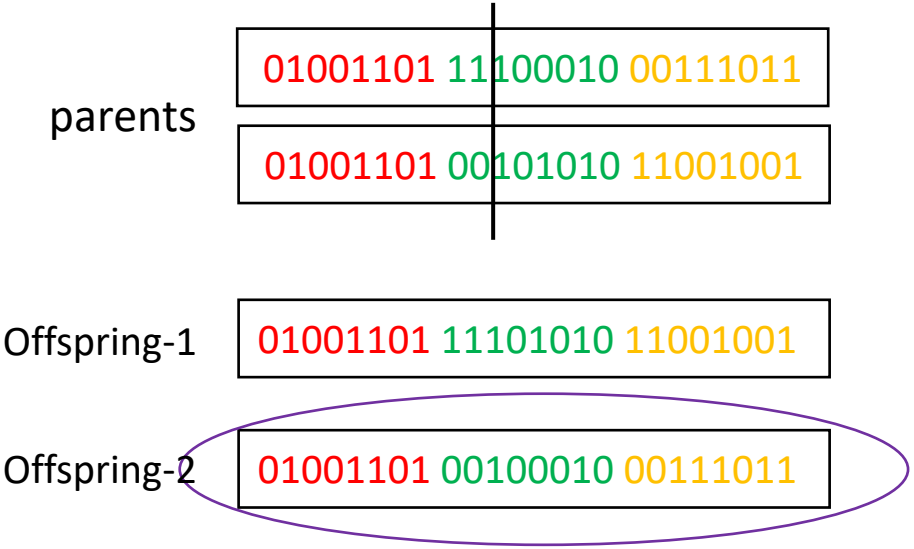


Swap mutation

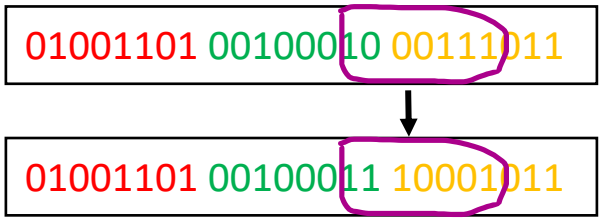


How to apply?

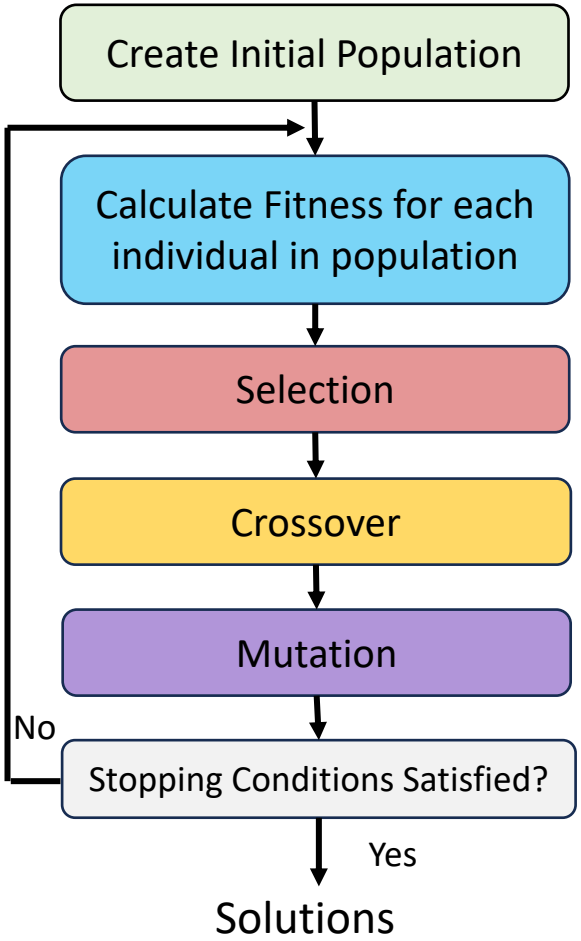


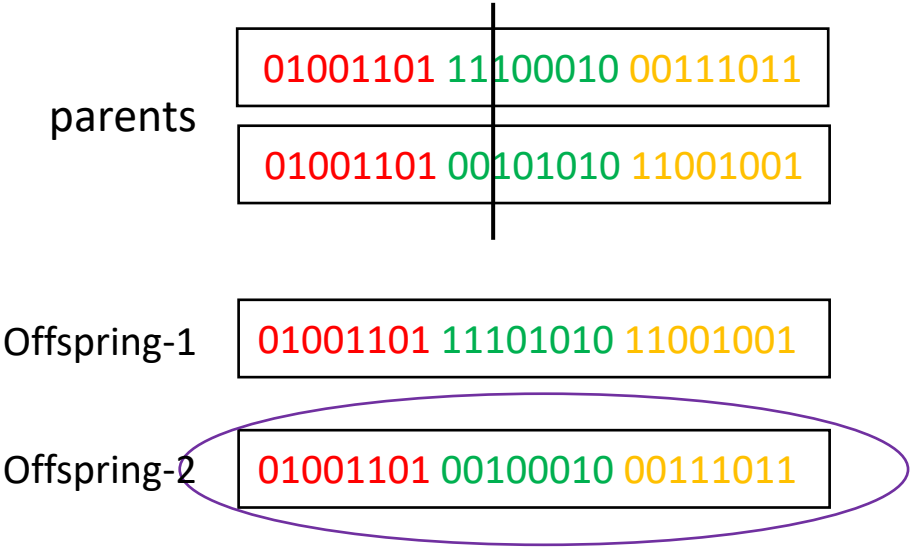


Inversion mutation

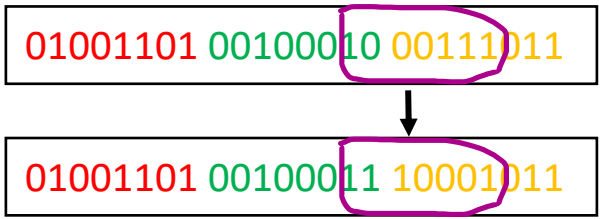


How to apply?





Inversion mutation



Crossover frequency:

All the time (100%)
All offsprings made via crossover

Never (0%)
Copy parents

It's reasonable to copy "some" chromosomes into the next generation

Mutation frequency:

Never (0%)
No change in copies/offsprings

Too often (~50%)
Huge variability preventing convergence

Rarely (~1%)
Additional diversity contributing to good solution

When to stop the evolution?

1. Max number of generation
2. Min level of diversity
3. Certain level of fitness
4. Certain number of generations when no significant fitness change happens

Encodings in Genetic Algorithm

Binary encoding

It is one of the commonest method of encoding and is widely used in genetic algorithms
The chromosome is represented as a string of 1 and 0

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \quad x, y, z \in [-3, 3]$$

x	y	z
01001101	11100010	00111011

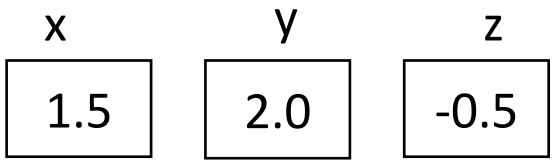
Chromosome: [01001101 11100010 00111011]

Encodings in Genetic Algorithm

Value encoding

- Represent a gene as some value
- Value can be an integer, real number, character or some object
- Use direct representations of the variables/design parameters
- Avoids any intermediate encoding and decoding steps
- The chromosome is represented as a string of 1 and 0

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \quad x, y, z \in [-3, 3]$$

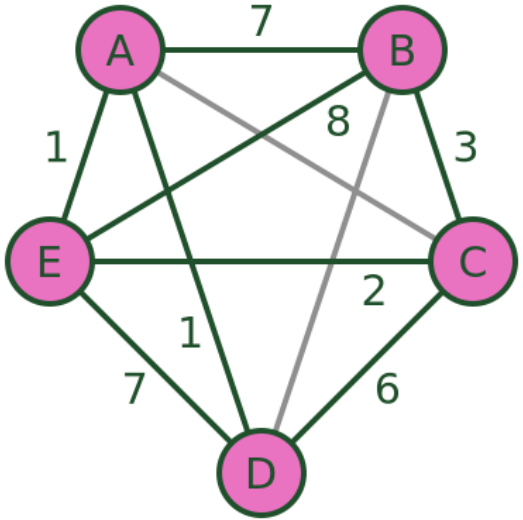


Chromosome: [1.5, 2.0, -0.5]

Encodings in Genetic Algorithm

Permutation (or Order) encoding

It is useful for order problems where the order of processing is particularly important

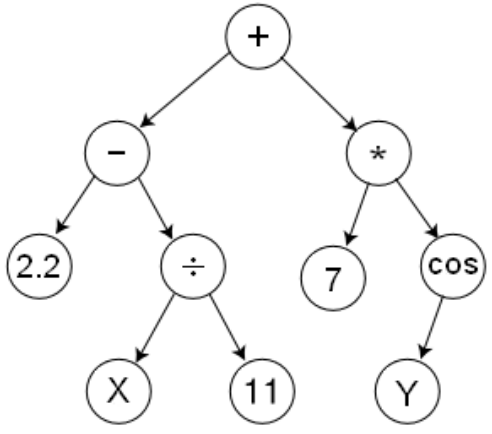


Example route: A→B→C→D→E→(back to A)
Encoding: [7, 3, 6, 7, 1]

Chromosome

Encodings in Genetic Algorithm

Tree/structure encoding



$$\left(2.2 - \left(\frac{X}{11}\right)\right) + \left(7 * \cos(Y)\right)$$

Genetic programming (GP)

Each node is an operation, and the leaves are variables or constants

This whole tree = ONE chromosome

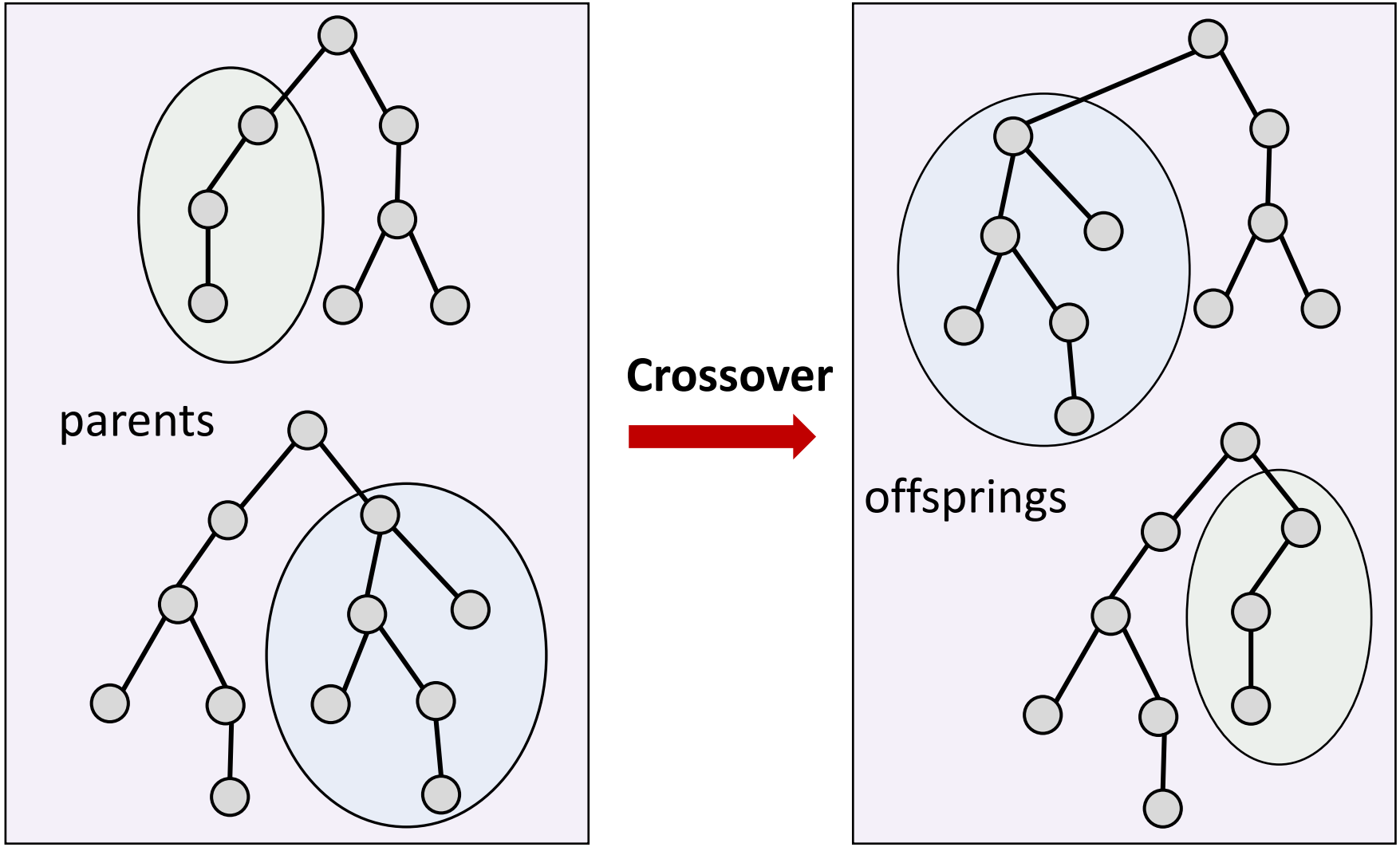
Genetic programming (GP)

$$f(x, y, z) = x^2 + y^2 + z^2 + \sin(4x)\cos(3y)\sin(2z) \qquad x, y, z \in [-3, 3]$$

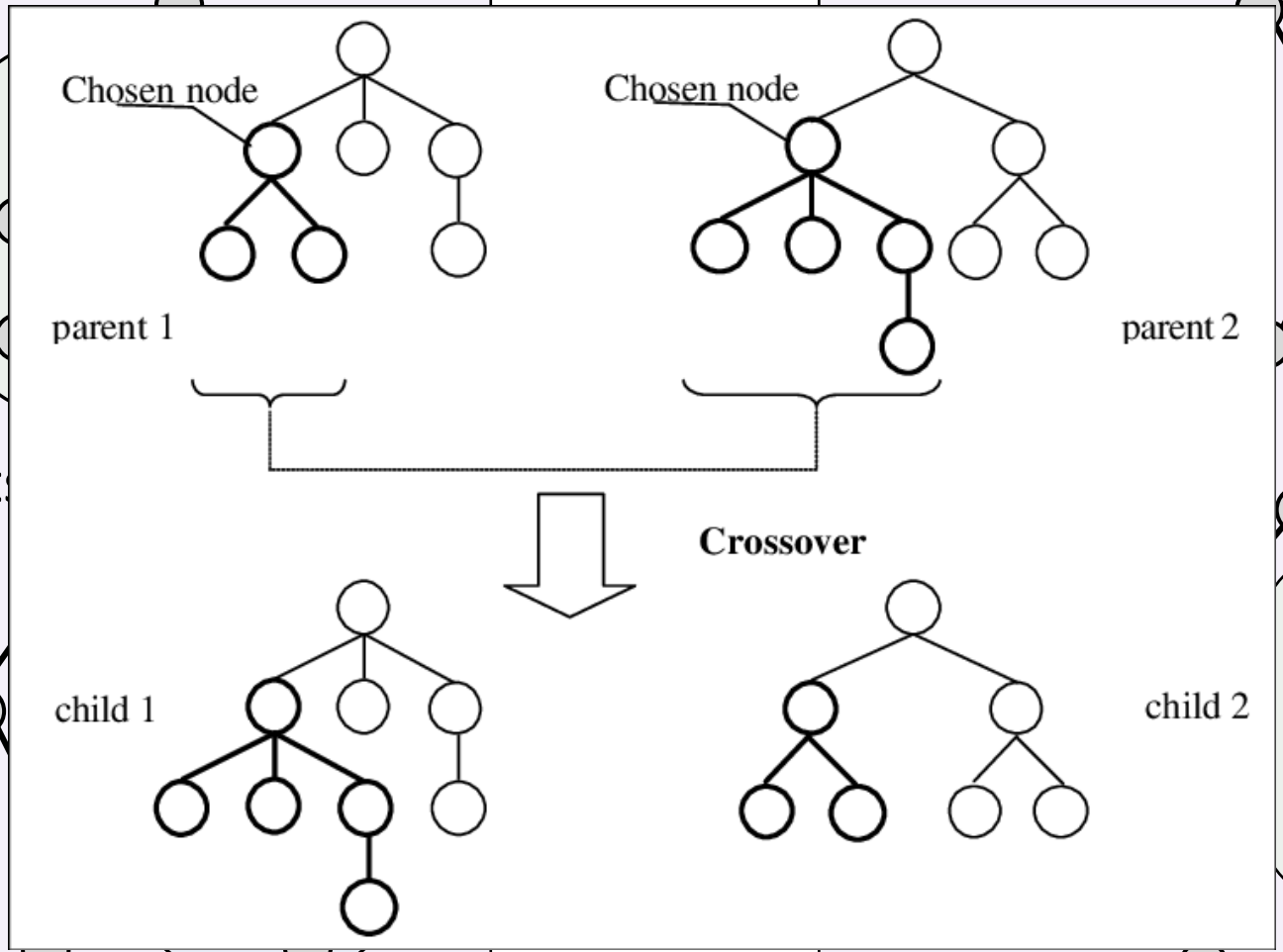
What are the best parameter values?

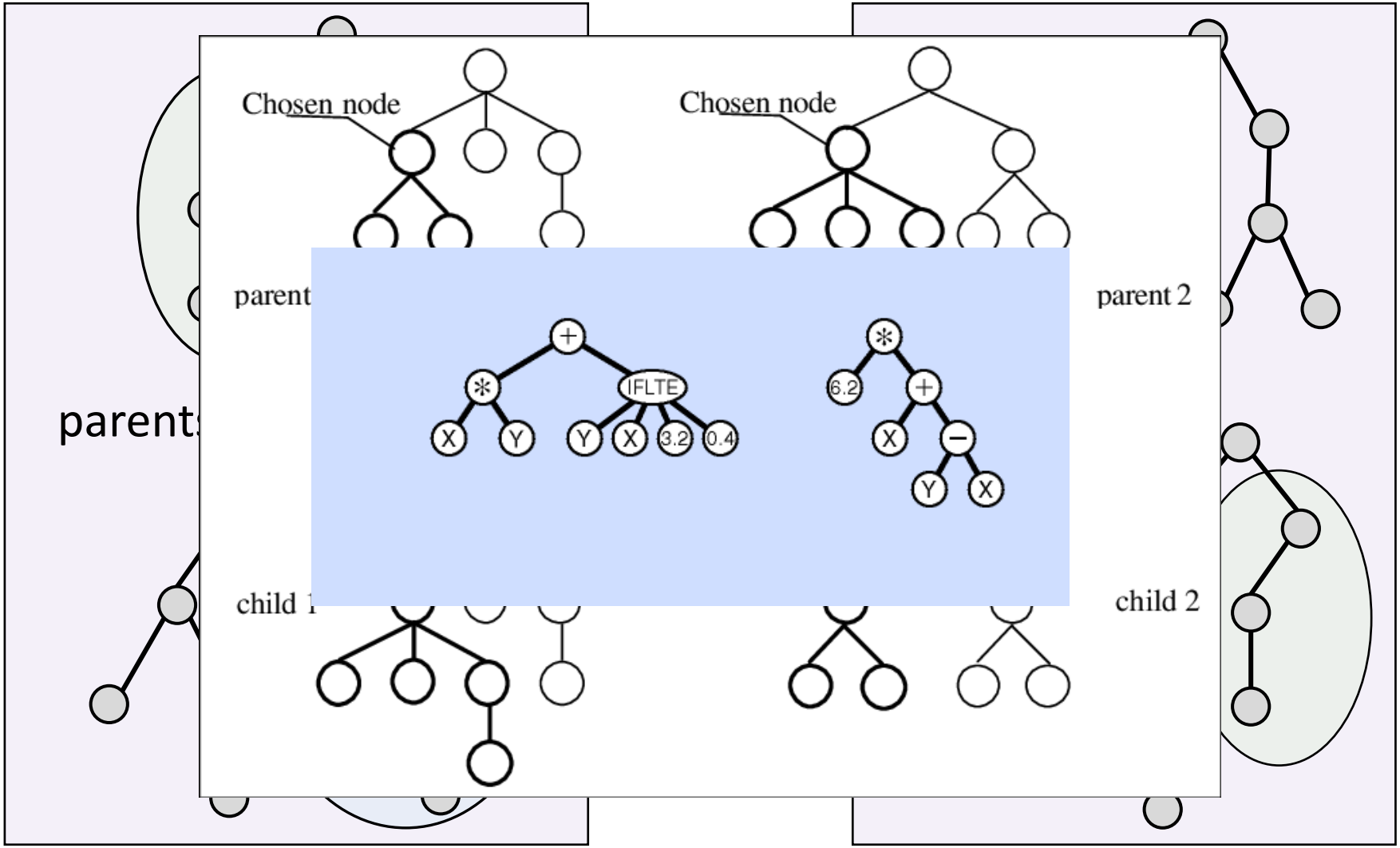
What should the formula/program/strategy be?

Structure, Not Just Parameters

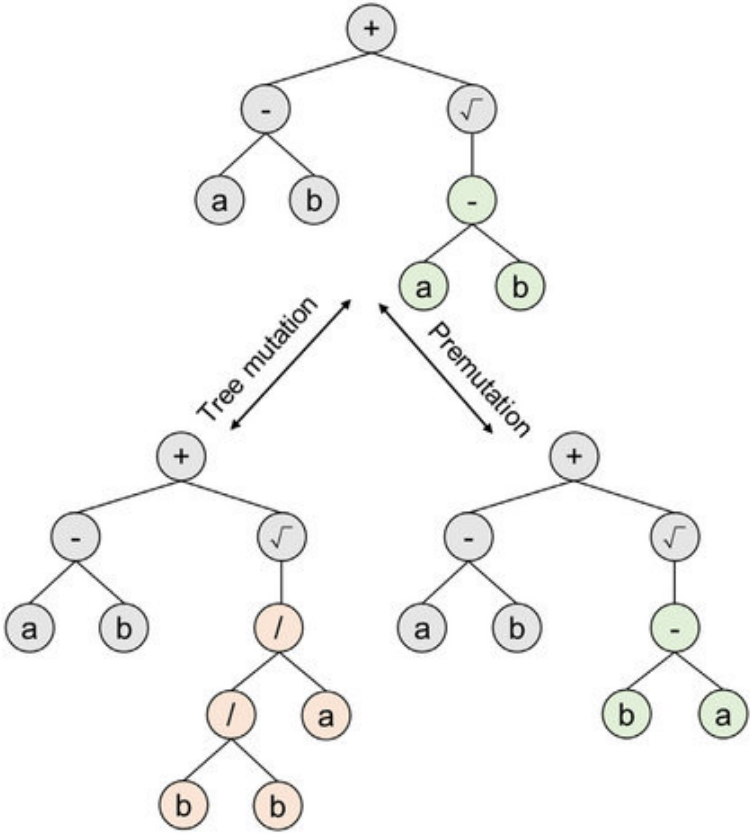

parents
Crossover
offsprings

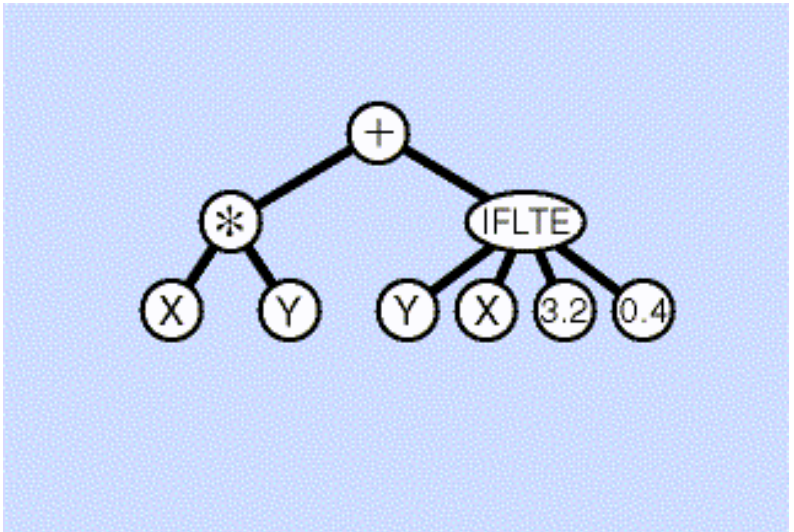
parents


Chosen node
Chosen node
parent 1
parent 2
Crossover
child 1
child 2


Chosen node
Chosen node
parent
parent 2
IFLTE
child 1
child 2

Genetic programming (GP)





Genetic programming (GP)

We have measurements:

$x = 1 \rightarrow y = 3$
$x = 2 \rightarrow y = 6$
$x = 3 \rightarrow y = 9$
$x = 4 \rightarrow y = 12$
$x = 5 \rightarrow y = 15$

$y = 3x$

Question: What is the formula? y = ???

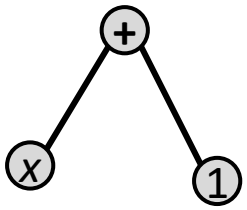
Genetic programming (GP)

Building Blocks (like LEGO pieces)

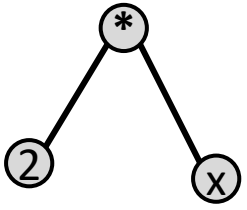
Functions (operators)

+ (addition)
- (subtraction)
* (multiplication)
/ (division)

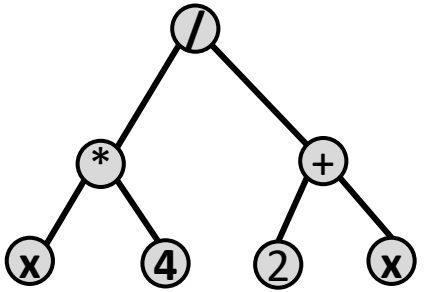
Terminals (values):
x (the input variable)
Constants: 1, 2, 3, 4, 5



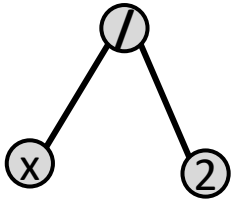
Formula: y = x + 1



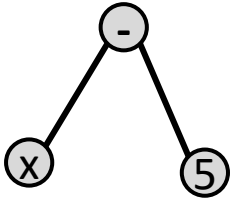
Formula: y = 2*x



Formula: y = x*4 / (2+x)

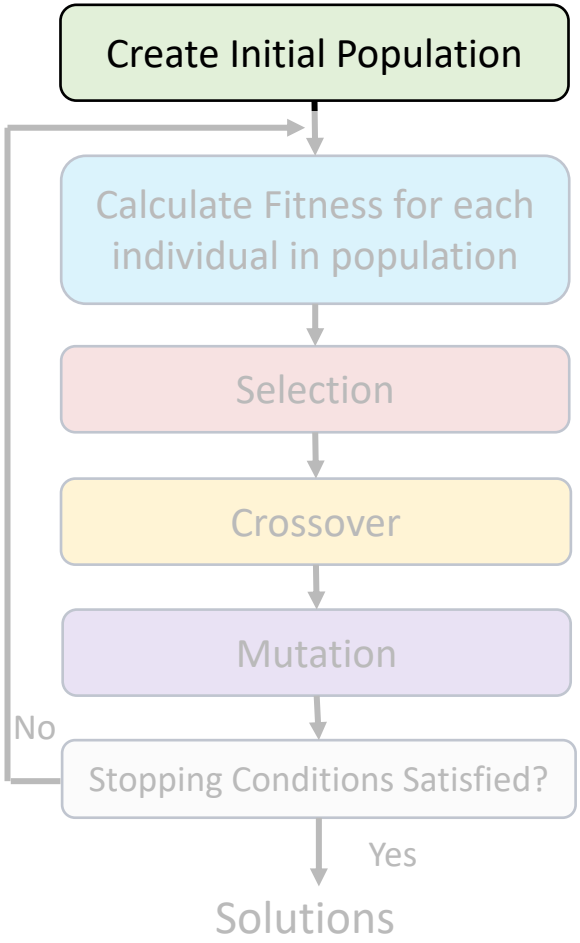


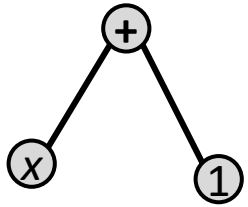
Formula: y = x/2



Formula: y = x-5

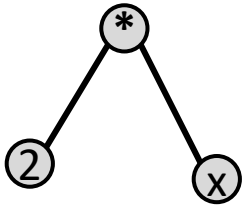
Genetic programming (GP)





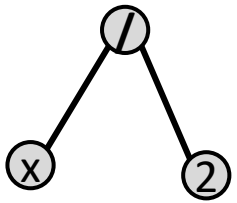
No.1

y = x + 1



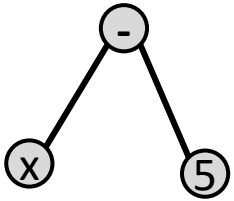
No.2

y = 2*x



No.3

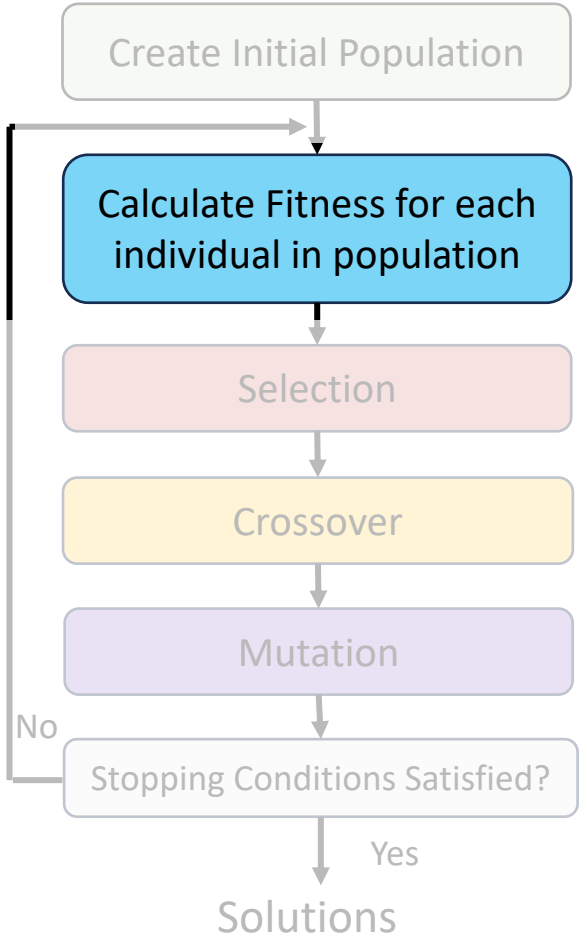
y = x/2

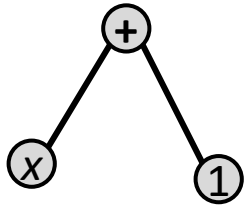


No.4

y = x-5

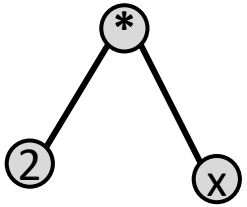
Genetic programming (GP)





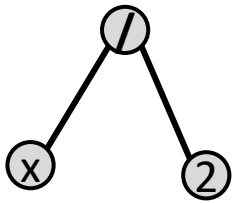
No.1

y = x + 1



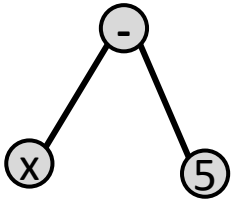
No.2

y = 2*x



No.3

y = x/2

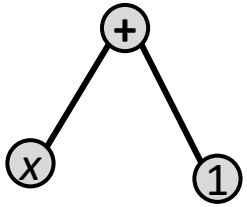


No.4

y = x-5

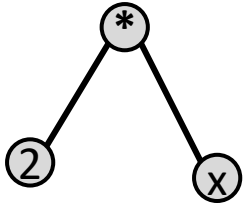
x	Target	Predicted	Error	Error²
1	3	1+1=2	3-2=1	1
2	6	2+1=3	6-3=3	9
3	9	3+1=4	9-4=5	25
4	12	4+1=5	12-5=7	49
5	15	5+1=6	15-6=9	81

Genetic programming (GP)



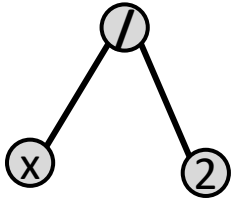
No.1

y = x + 1



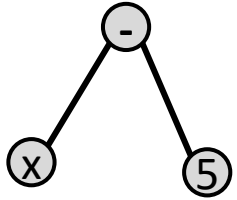
No.2

y = 2*x



No.3

y = x/2



No.4

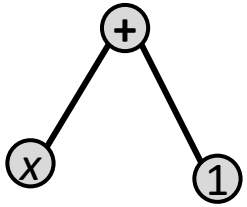
y = x-5

x	Target	Predicted	Error	Error²
1	3	1+1=2	3-2=1	1
2	6	2+1=3	6-3=3	9
3	9	3+1=4	9-4=5	25
4	12	4+1=5	12-5=7	49
5	15	5+1=6	15-6=9	81

Total Squared Error = 1 + 9 + 25 + 49 + 81 = 165

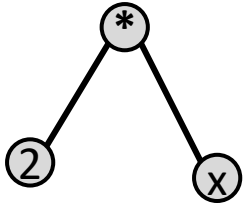
Fitness = 1/(1 + 165) = 0.006

Genetic programming (GP)



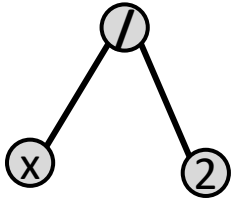
No.1

y = x + 1



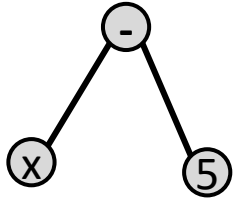
No.2

y = 2*x



No.3

y = x/2



No.4

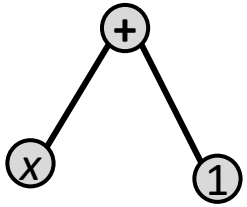
y = x-5

x	Target	Predicted	Error	Error²
1	3	2×1=2	3-2=1	1
2	6	2×2=4	6-4=2	4
3	9	2×3=6	9-6=3	9
4	12	2×4=8	12-8=4	16
5	15	2×5=10	15-10=5	25

Total Squared Error = 1 + 4 + 9 + 16 + 25 = 55

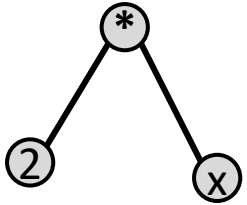
Fitness = 1/(1 + 55) = 0.018

Genetic programming (GP)



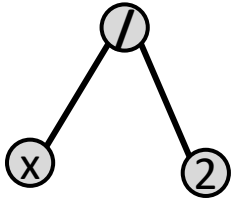
No.1

y = x + 1



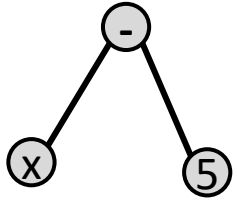
No.2

y = 2*x



No.3

y = x/2



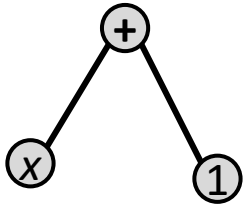
No.4

y = x-5

x	Target	Predicted	Error	Error²
1	3	1-5=-4	3-(-4)=7	49
2	6	2-5=-3	6-(-3)=9	81
3	9	3-5=-2	9-(-2)=11	121
4	12	4-5=-1	12-(-1)=13	169
5	15	5-5=0	15-0=15	225

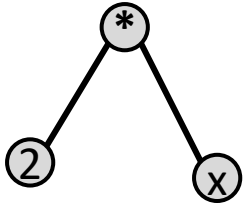
Total Squared Error = 645
Fitness = 1/(1 + 645) = 0.0015

Genetic programming (GP)



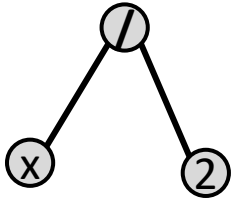
No.1

y = x + 1



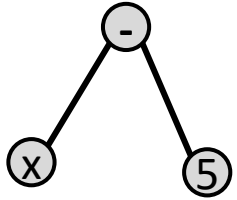
No.2

y = 2*x



No.3

y = x/2



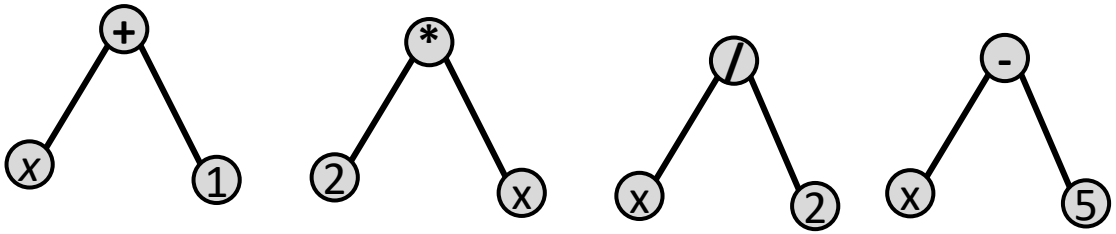
No.4

y = x-5

x	Target	Predicted	Error	Error²
1	3	1÷2=0.5	2.5	6.25
2	6	2÷2=1	5	25
3	9	3÷2=1.5	7.5	56.25
4	12	4÷2=2	10	100
5	15	5÷2=2.5	12.5	156.25

Total Squared Error = 343.75
Fitness = 1/(1 + 343.75) = 0.0029

Genetic programming (GP)



No.1

y = x + 1

No.2

y = 2*x

No.3

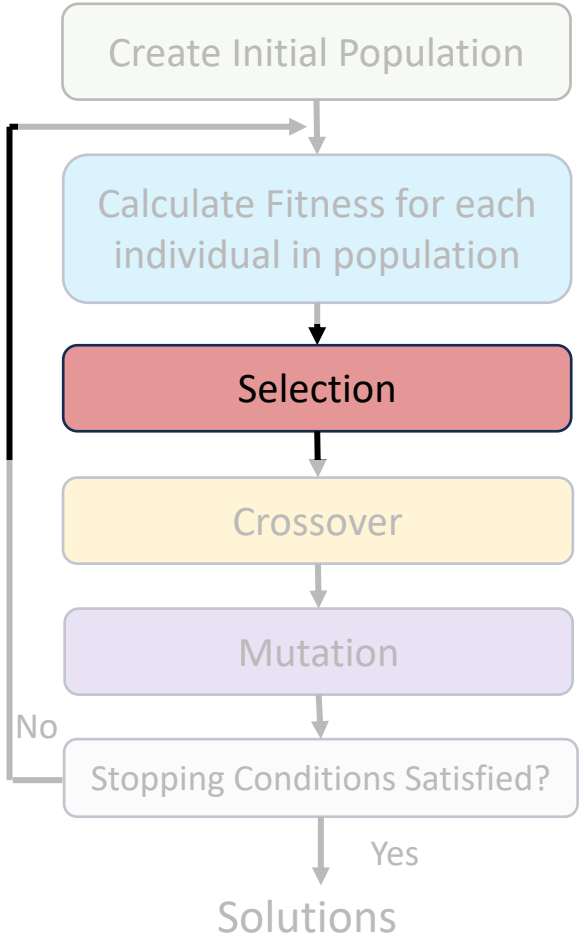
y = x/2

No.4

y = x-5

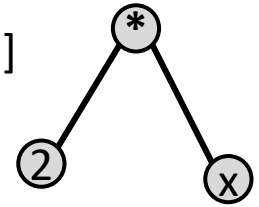
chromosome	Formula	Error	Fitness
Individual 2	y = 2*x	55	0.018
Individual 1	y = x+1	165	0.006
Individual 4	y = x÷2	343.75	0.003
Individual 3	y = x-5	645	0.002

Genetic programming (GP)

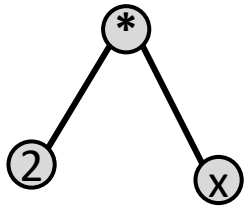


Tournament Selection (size = 2)

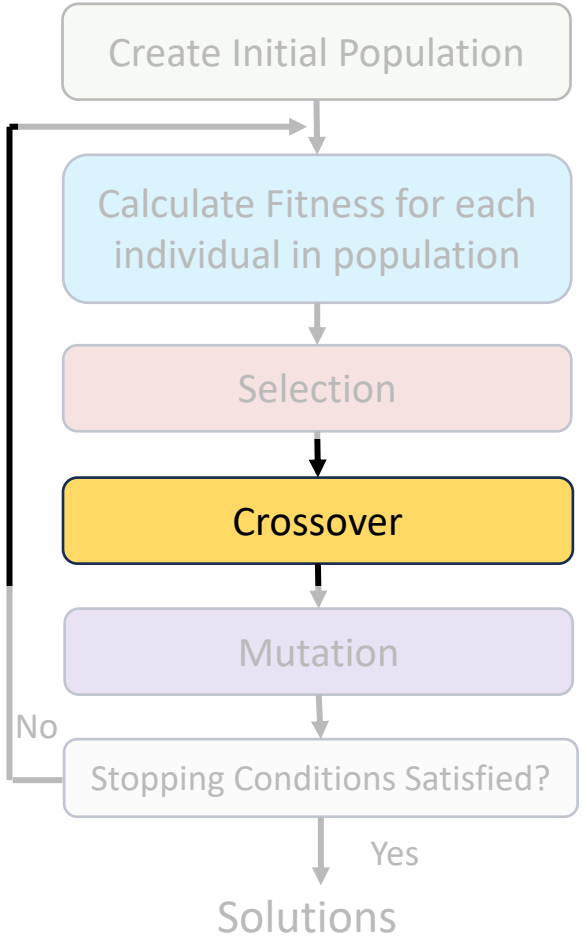
Tournament 2: Pick 2 random individuals
[Individual 2 (0.018), Individual 4 (0.003)]
Winner: Individual 2 ✓



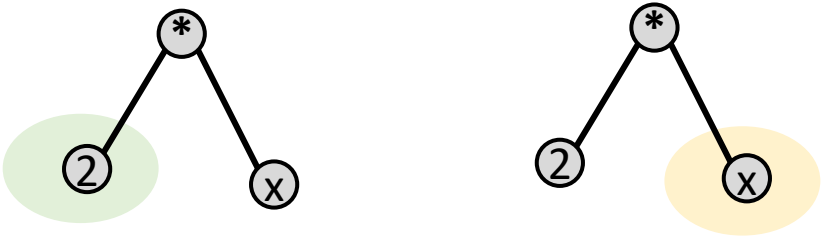
Tournament 2: Pick 2 random individuals
[Individual 1 (0.006), Individual 2 (0.018)]
Winner: Individual 2 ✓



Genetic programming (GP)



parents



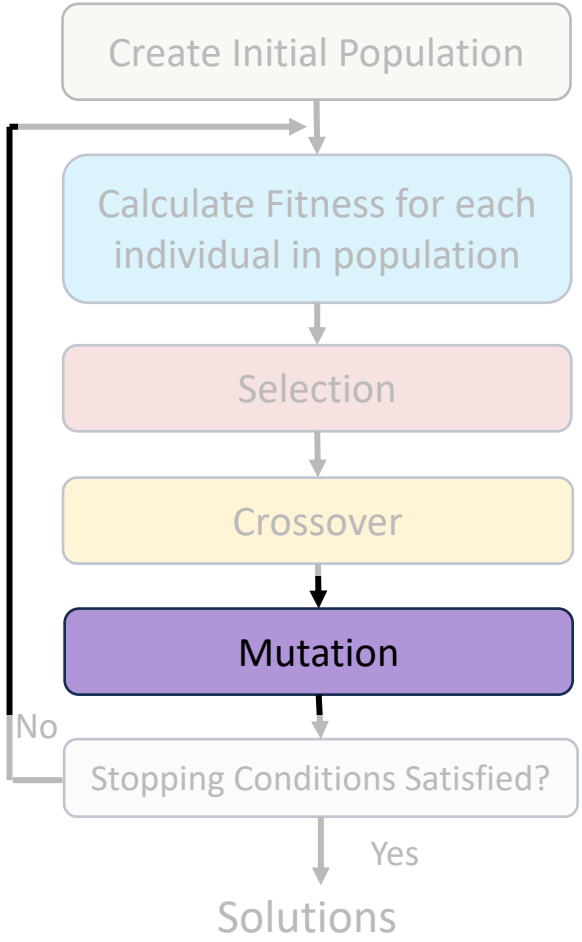
offsprings



y = x *x

y = 2*2

Genetic programming (GP)

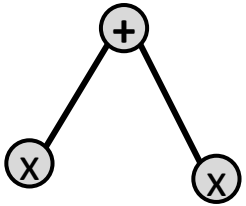


offsprings

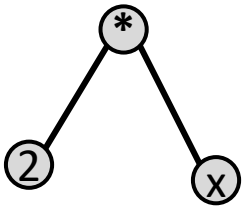


Random Changes, Change one node randomly

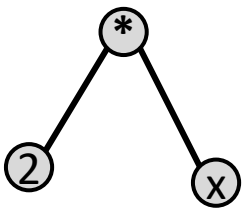
* to +

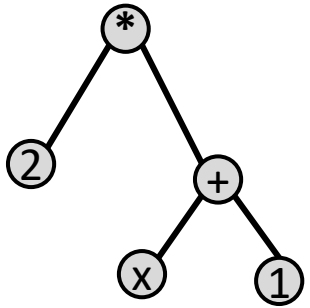


2 to x

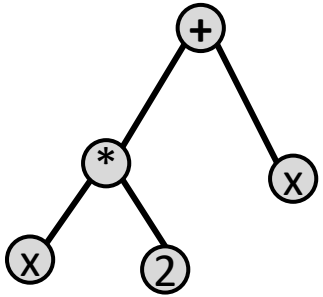


Genetic programming (GP)





...



y = 3x

Symbolic regression

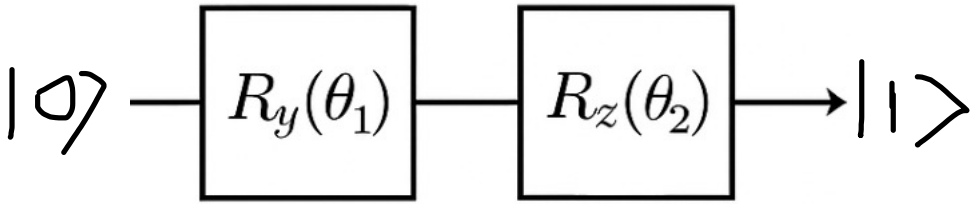
x = 1 → y = 3
x = 2 → y = 6
x = 3 → y = 9
x = 4 → y = 12
x = 5 → y = 15

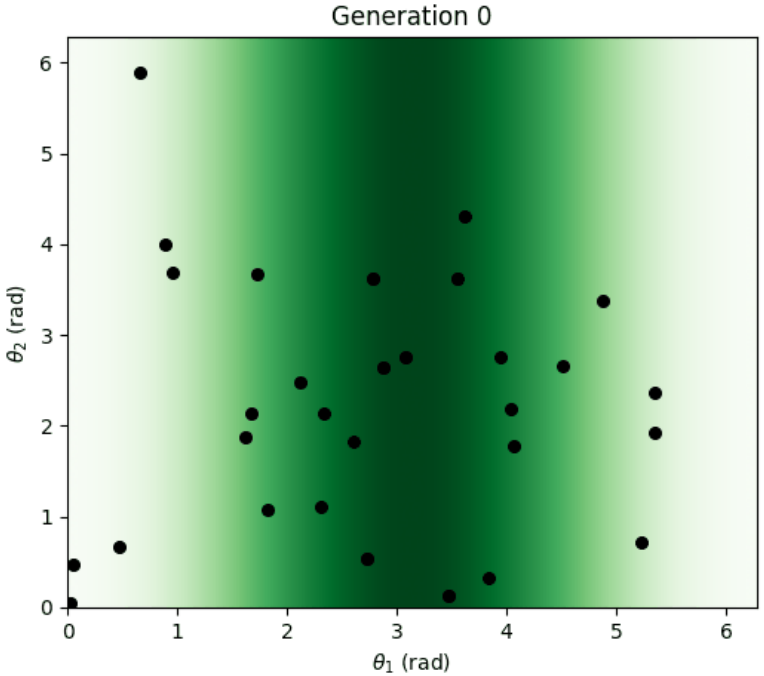
Question: What is the formula? y = ???

Examples in (Quantum) Physics

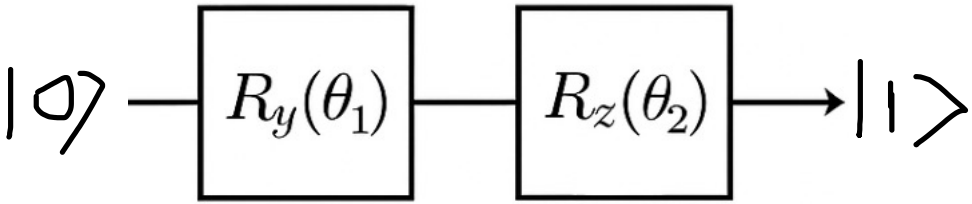
$$R_y(\theta) := e^{-i\theta\sigma_y/2} = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

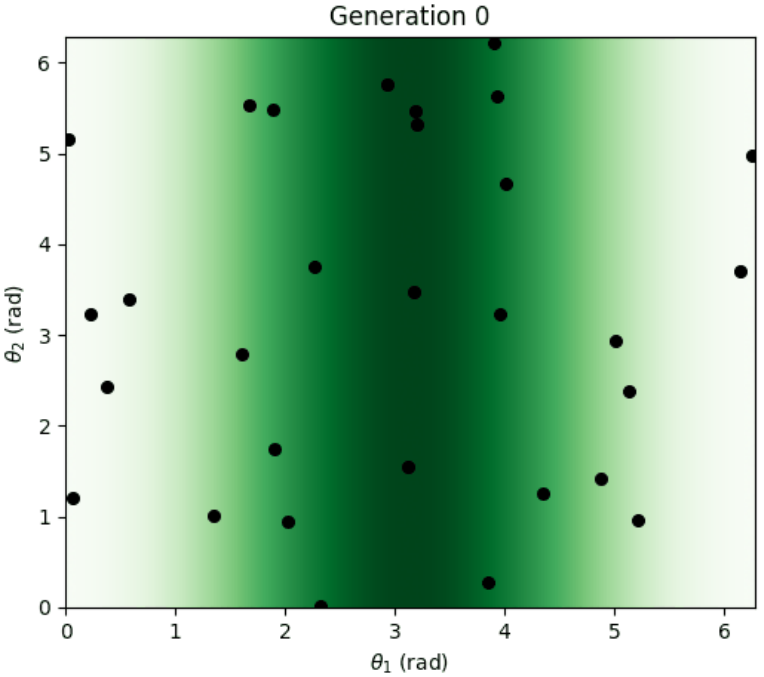
$$R_z(\theta) := e^{-i\theta\sigma_z/2} = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$





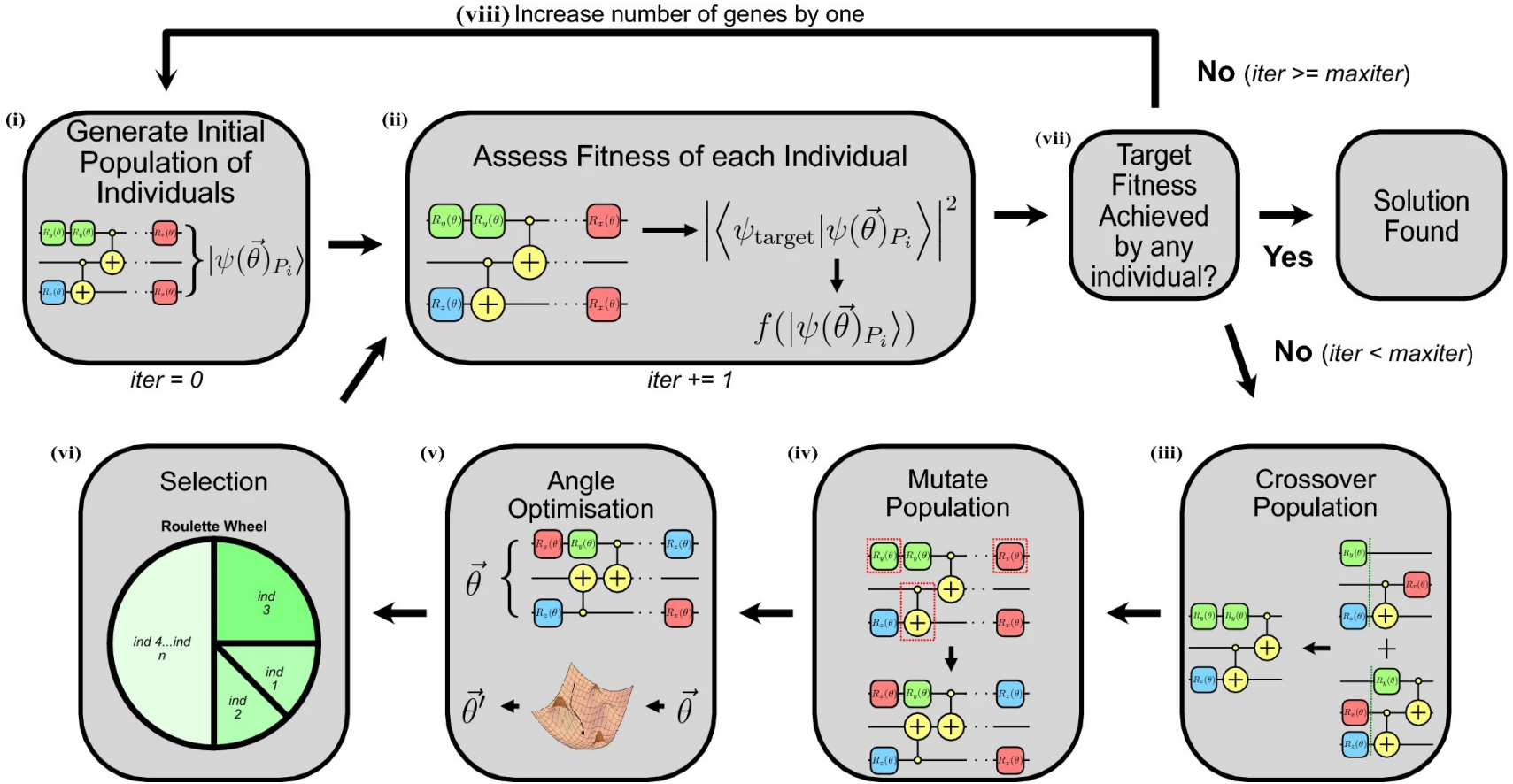
Examples in (Quantum) Physics



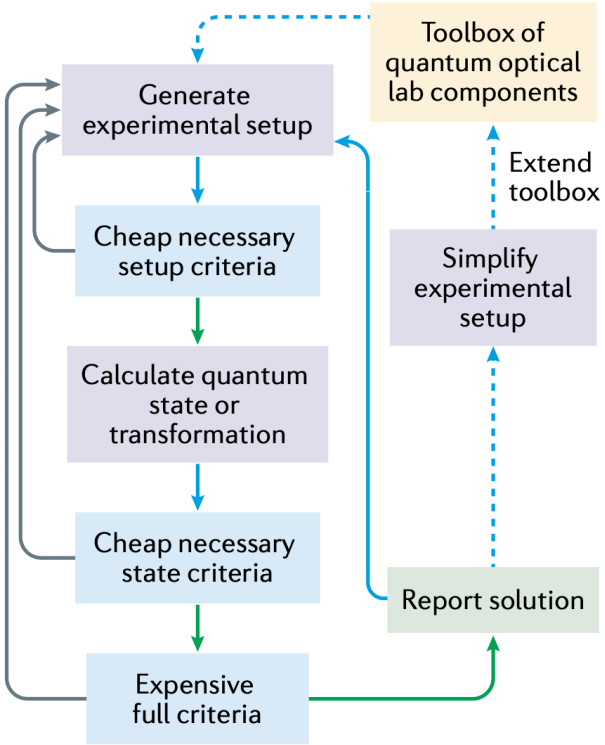


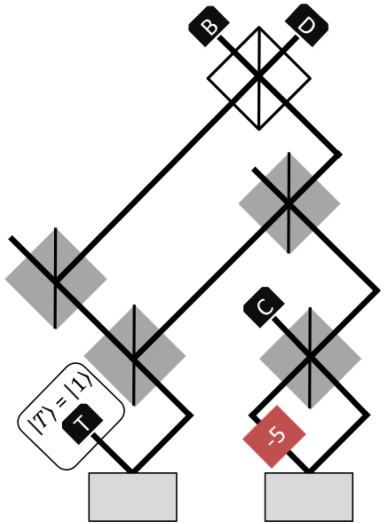
state preparation on quantum computers

Creevey, F.M., Hill, C.D. & Hollenberg, L.C.L.. *Sci Rep* **13**, 11956 (2023).



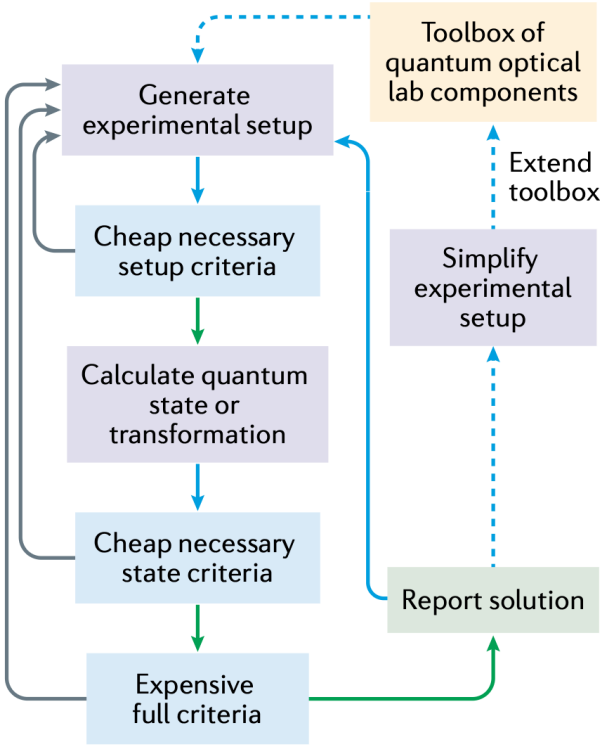
Examples in (Quantum) Physics

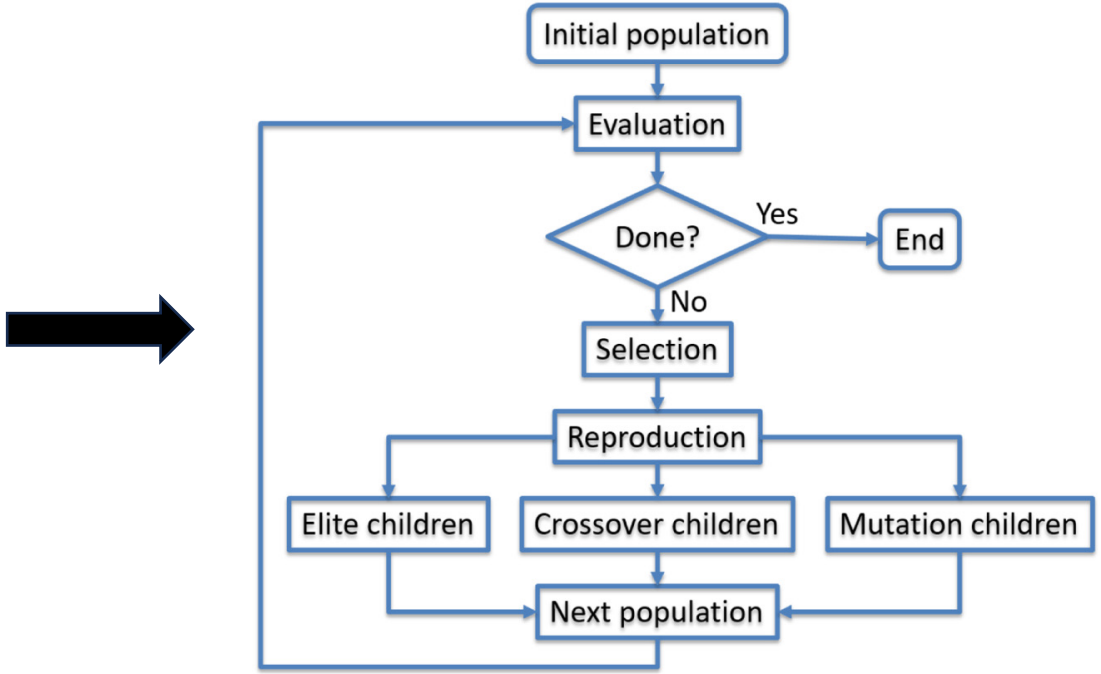




Rosanna Nichols *et al* 2019 *Quantum Sci. Technol.* **4** 045012

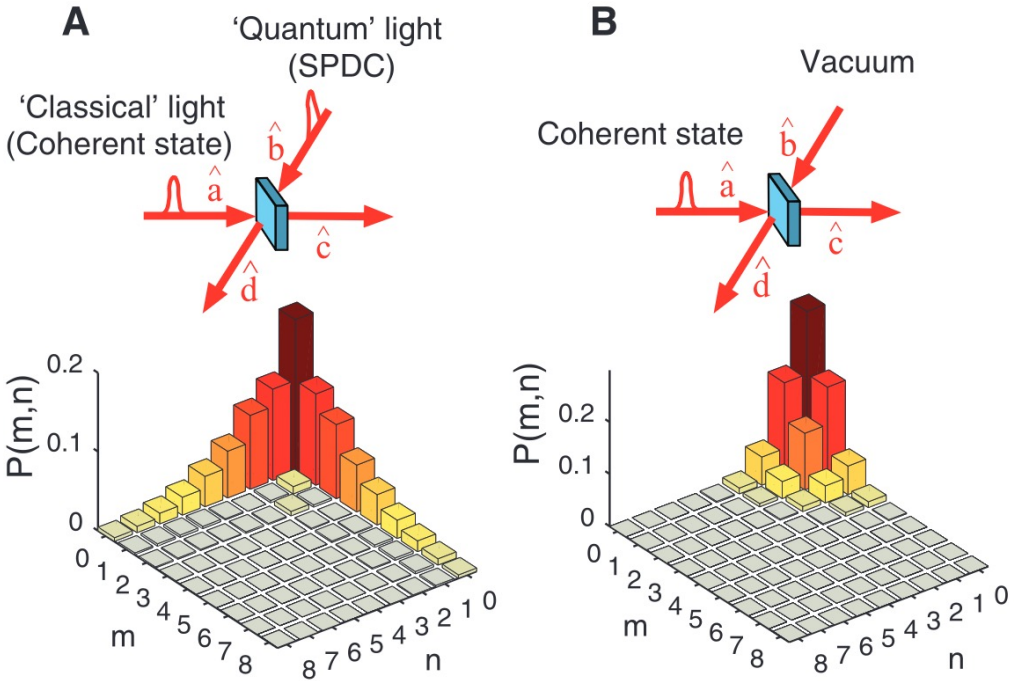
Examples in (Quantum) Physics





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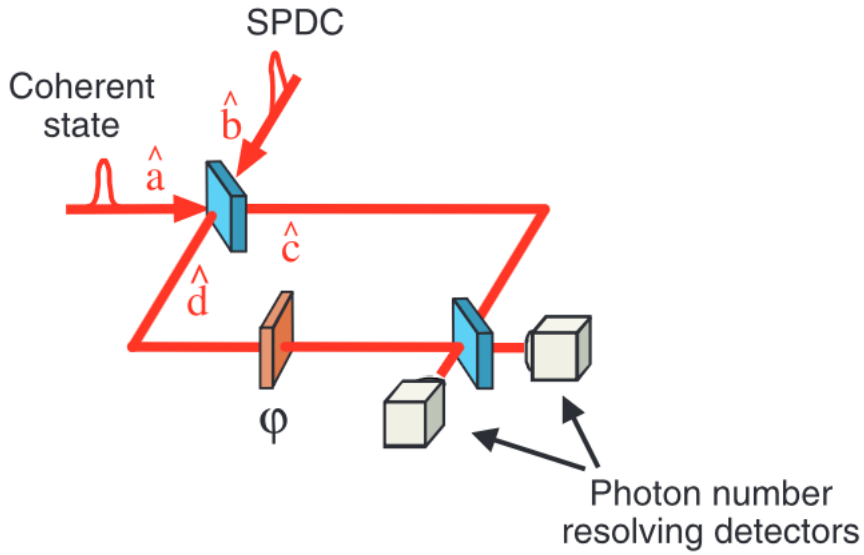
Examples in (Quantum) Physics

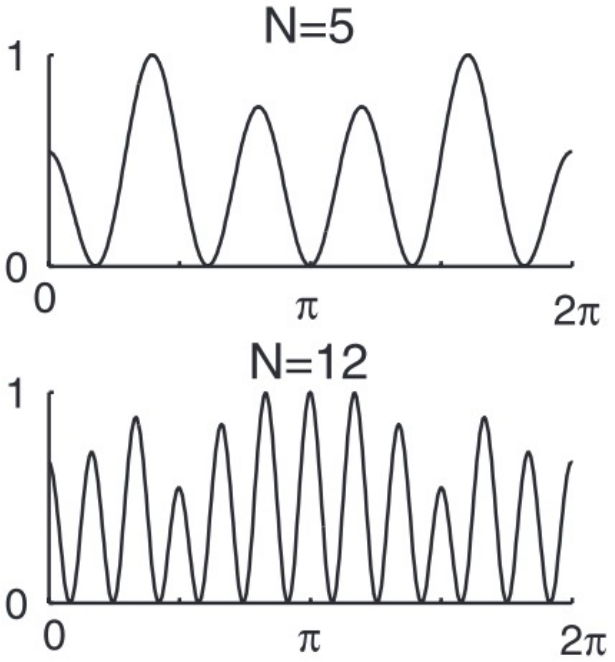


$$|N::0\rangle_{A,B} \equiv \frac{1}{\sqrt{2}}\left(|N,0\rangle_{A,B} + |0,N\rangle_{A,B}\right)$$

Itai Afek *et al.*,High-NOON States by Mixing Quantum and Classical Light.*Science***328**,879-881(2010)

Examples in (Quantum) Physics

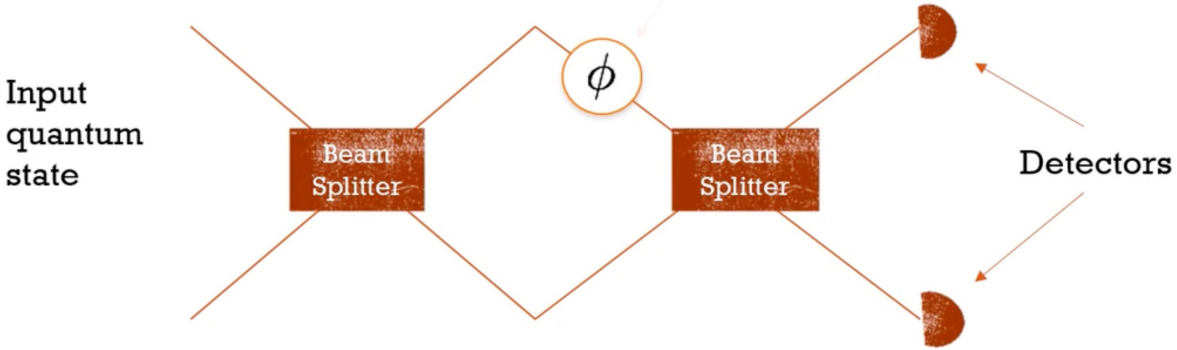


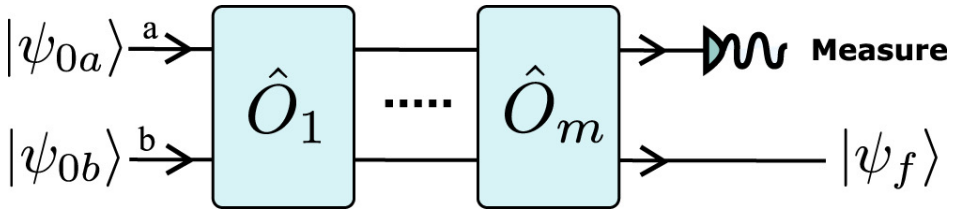


Itai Afek *et al.*,High-NOON States by Mixing Quantum and Classical Light.*Science***328**,879-881(2010)

Examples in (Quantum) Physics

- Quantum metrology: measure a phase difference between two paths in an interferometer





$$\hat{D}(\alpha) = \exp(\alpha\hat{a}^\dagger - \alpha^*\hat{a})$$

$$|\alpha\rangle = \exp(-|\alpha|^2/2)\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle$$

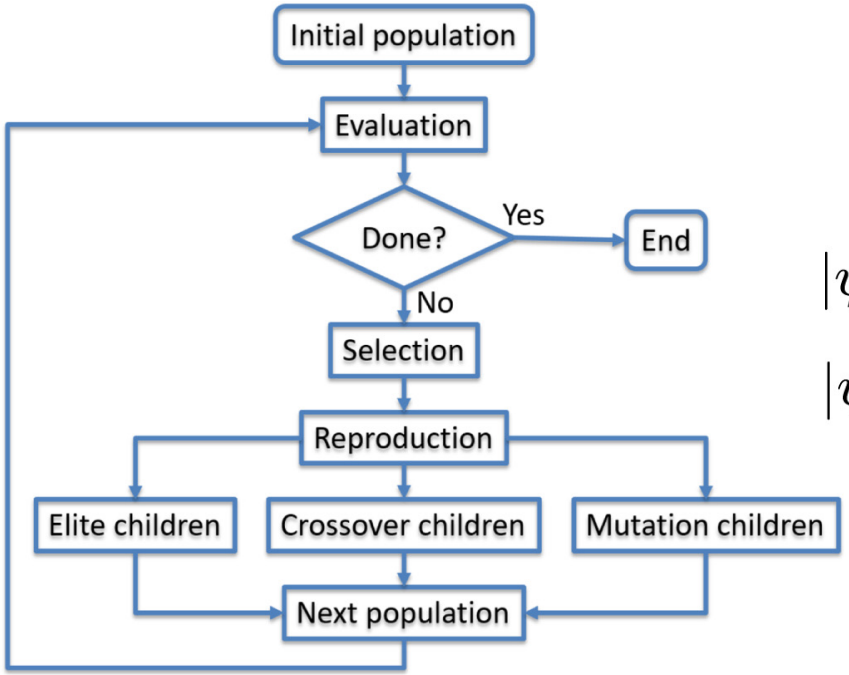
$$\hat{S}_i(\zeta) = \exp\left[\frac{1}{2}(\zeta^*\hat{a}_i^2 - \zeta\hat{a}_i^{\dagger 2})\right],$$

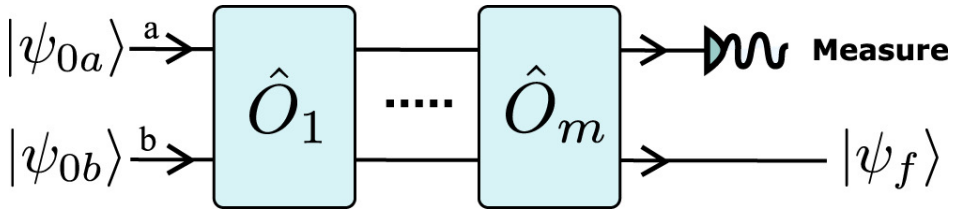
$$\hat{S}_{ij}(\zeta) = \exp\left[\zeta^*\hat{a}_i\hat{a}_j - \zeta\hat{a}_i^\dagger\hat{a}_j^\dagger\right]$$

$$\hat{U}_{ij} = \exp\left[\theta(\hat{a}_i\hat{a}_j^\dagger - \hat{a}_i^\dagger\hat{a}_j)\right]$$

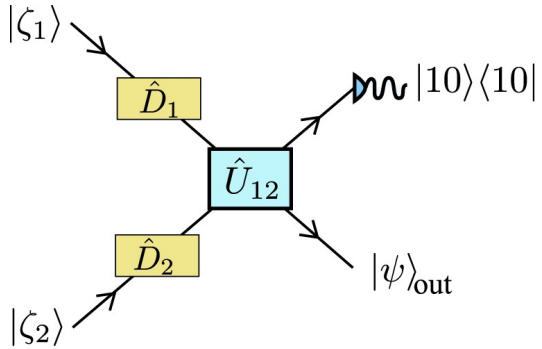
Rosanna Nichols *et al* 2019 *Quantum Sci. Technol.* **4** 045012

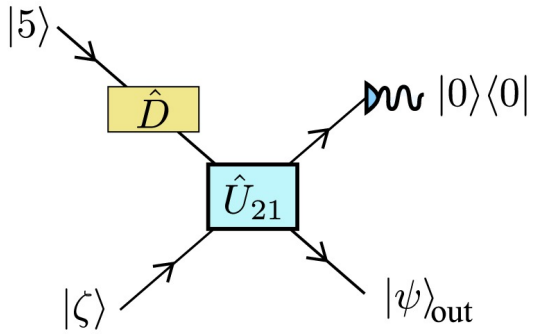
Examples in (Quantum) Physics

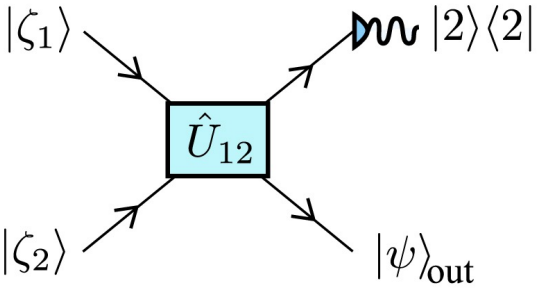


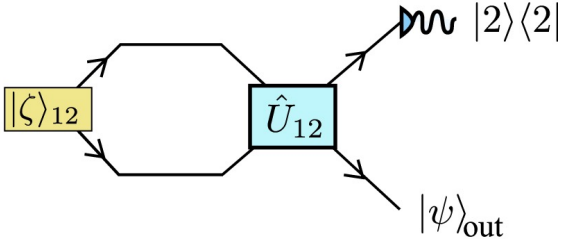


Examples in (Quantum) Physics



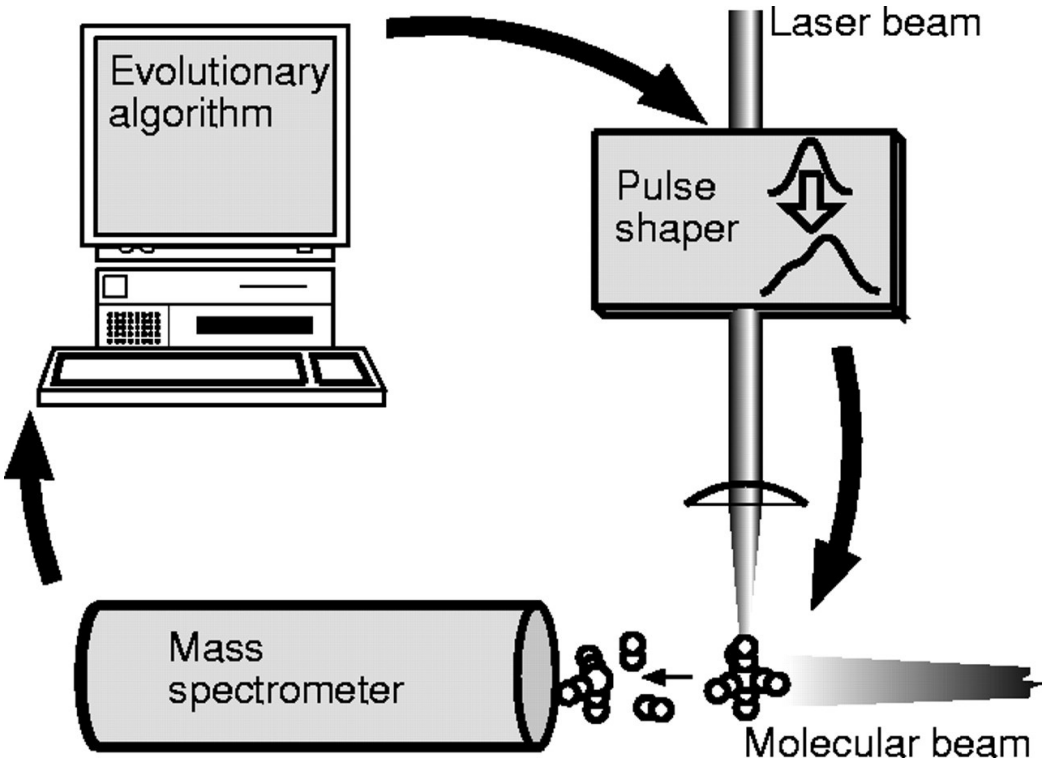






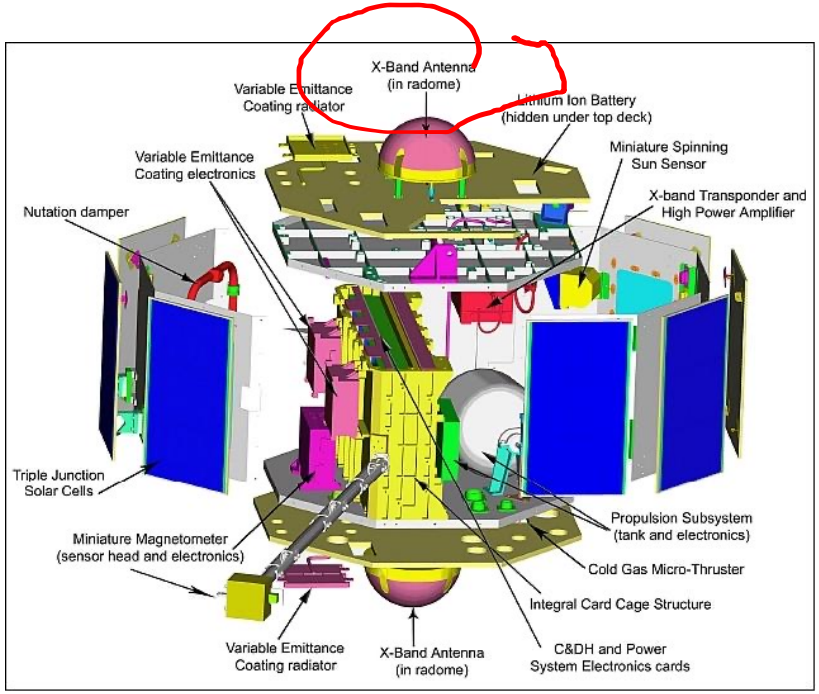
Rosanna Nichols *et al* 2019 *Quantum Sci. Technol.* **4** 045012

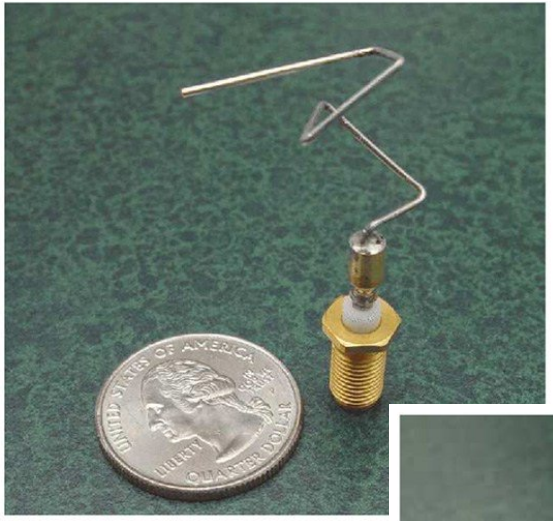
Teaching Lasers to Control Molecules



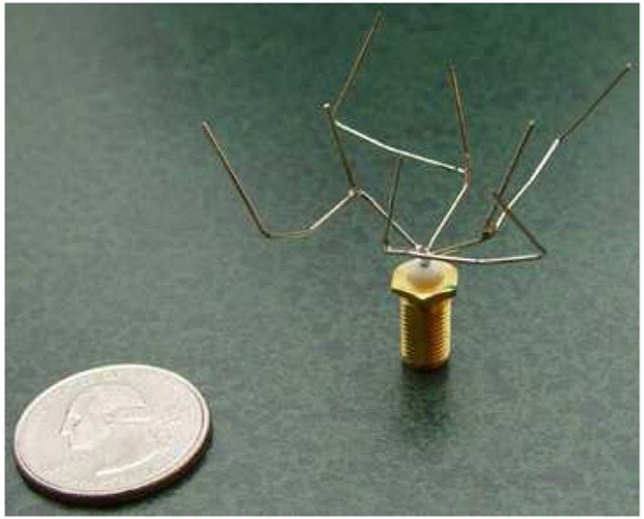
Judson, R. S., and H. Rabitz, Phys.Rev.Lett.68, 1500 (1992)
A. Assion *et al., .Science*282,919-922 (1998)

Examples





ST5 satellite antenna



Hornby, Gregory, et al. "Automated antenna design with evolutionary algorithms." *Space 2006*

ST5 satellite antenna

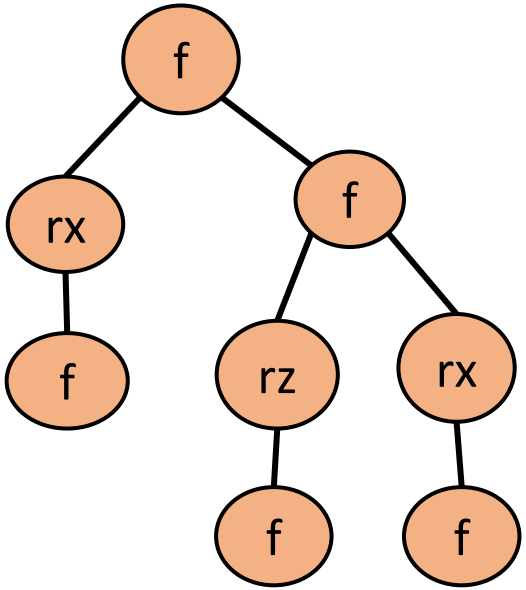
Genotype is a tree-structured encoding that specifies the construction of a wire form
Genotype specifies design of 1 arm in 3-space

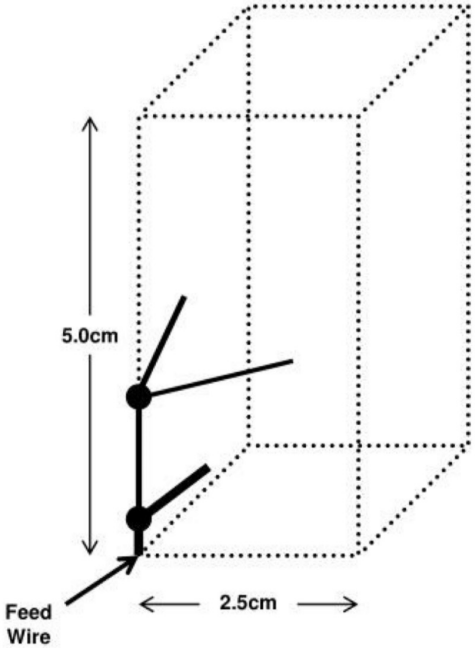
Commands:

- forward(length radius)
- rotate_x(angle)
- rotate_y(angle)
- rotate_z(angle)

Forward() command can have 0,1,2, or 3 children.

Rotate_x/y/z() commands have exactly 1 child (always non-terminal).





Hornby, Gregory, et al. "Automated antenna design with evolutionary algorithms." *Space 2006*

Different ST5 satellite antenna designs

